

Managerial Economics

Introduction

Understanding market demand

Understanding market forces – Demand,
supply, equilibrium

Term I

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Course Objective

- ▶ To provide the students an overview of *microeconomic* analysis required for managerial decision making in the domain of business.

The discipline of economic science

Economics

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graph TD; Economics[Economics] --> Microeconomics[Microeconomics]; Economics --> Macroeconomics[Macroeconomics]; Microeconomics --> MicroDescription[Helps in firm level strategic decision-making (choice of technology, finance, price, quantity, quality, incentives, so on)]; Macroeconomics --> MacroDescription[Helps understand Economic environment (Govt. policies, International economic linkages, Financial crisis)];
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Microeconomics

Macroeconomics

Helps in firm level strategic decision-making (choice of technology, finance, price, quantity, quality, incentives, so on)

Helps understand Economic environment (Govt. policies, International economic linkages, Financial crisis)

Role of managers

- ▶ Identify goals
- ▶ Identify constraints
- ▶ Take decision to achieve the goal in the most efficient manner

Fundamental questions of a firm

▶ **Product** selection

- Market selection and segmentation
- How competitive is the market?

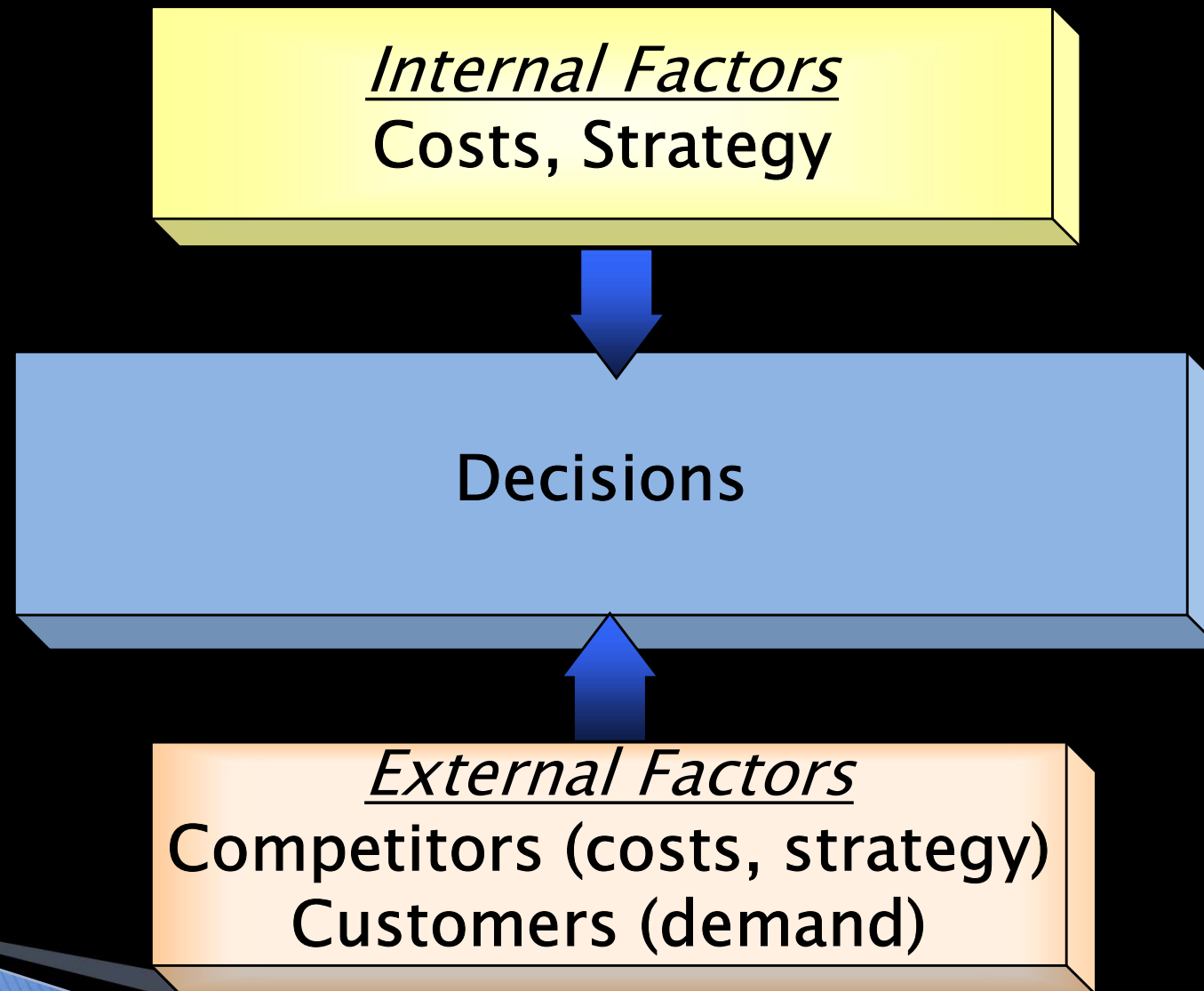
▶ **Process** of production

- Choice of technology – to produce at least cost

▶ **Price**

- How much to produce and what price to charge?
- Objective (generally) is profit maximization

Factors Affecting Microeconomic Decisions



What you need to understand at the microeconomic level?

- ▶ Behavior of (rational) individuals (*homo economicus*) – consumers, employees etc.
- ▶ Factors affecting firm behavior – production technology and costs
- ▶ Market forces and industry structure – revenue, contribution and profit
- ▶ Conflicts, incentives and strategic decision-making
- ▶ Role of information in business and possibility of market failure in absence of perfect information

Mathematics prerequisite

- ▶ Co-ordinate geometry (basic)
 - How do you geometrically represent an equation?
- ▶ Calculus (basic)
 - Functions
 - Derivatives
 - First order
 - Second order
 - Maxima and minima
- ▶ Probability and expectation (very basic)

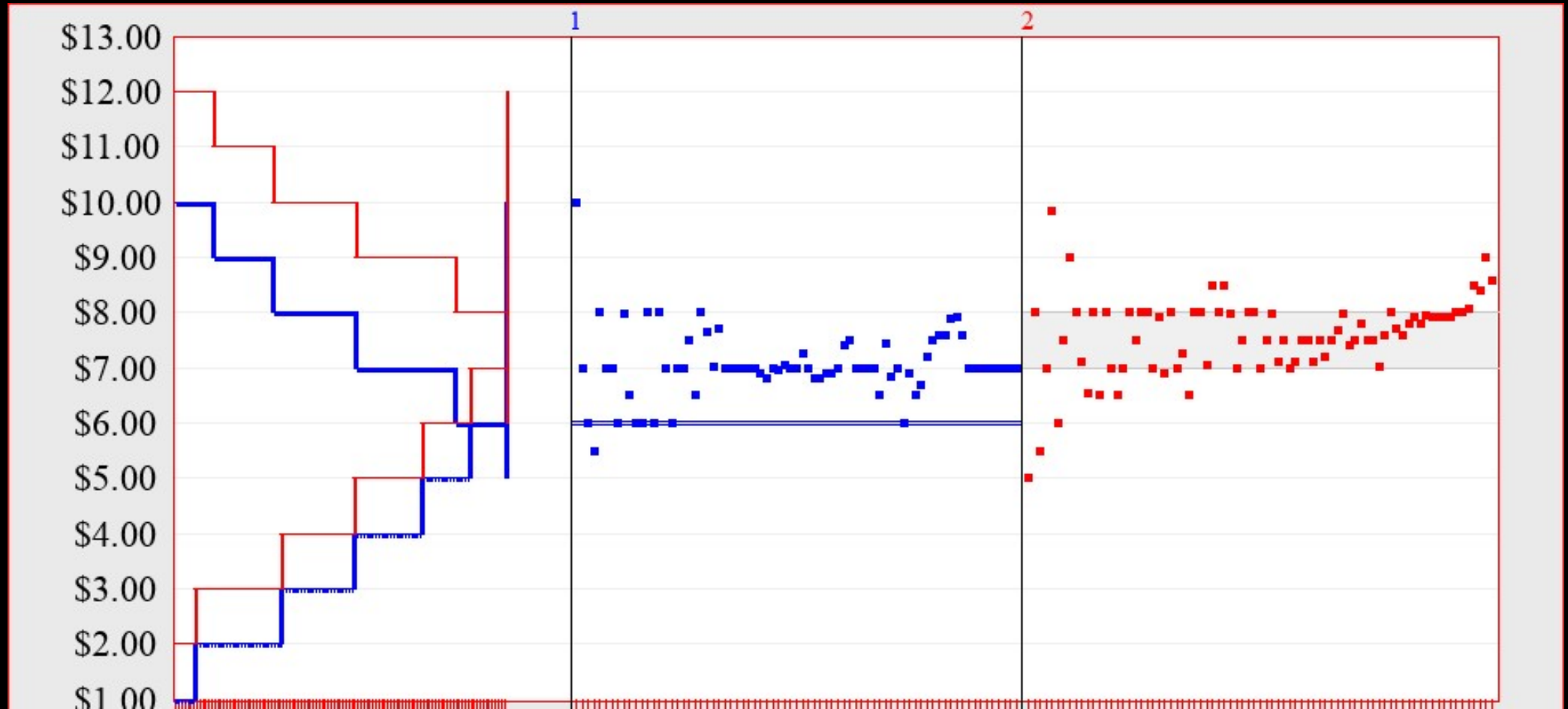
Evaluation schedule

- ▶ 2 quizzes – [2 x 30] 60 marks
 - Quiz 1 on 21 Sept
 - Quiz 2 on 12 Oct
- ▶ End-term – 40 marks

A market game

- ▶ The program allocates your role – Buyer or Seller. The role remained unchanged throughout.
- ▶ The program allocates the number of units of a hypothetical good you can buy or sell (depending on your role).
- ▶ The program allocates your value for different units if you are a Buyer, and your cost of producing different units of the good if you are a Seller.
- ▶ Buyers are not allowed to buy at a price higher than their value, and sellers are not allowed to sell at a price lower than their cost.
- ▶ Buyers will bid a price for the good and sellers will ask for a price.
- ▶ The program displays the highest bid price and the lowest ask price.
- ▶ When a bid price is greater than or equal to an ask price, a transaction will be done.
- ▶ Buyer's earning = $(\text{Value} - \text{Price})$ and seller's earning = $(\text{Price} - \text{Cost})$.

Market game outcome



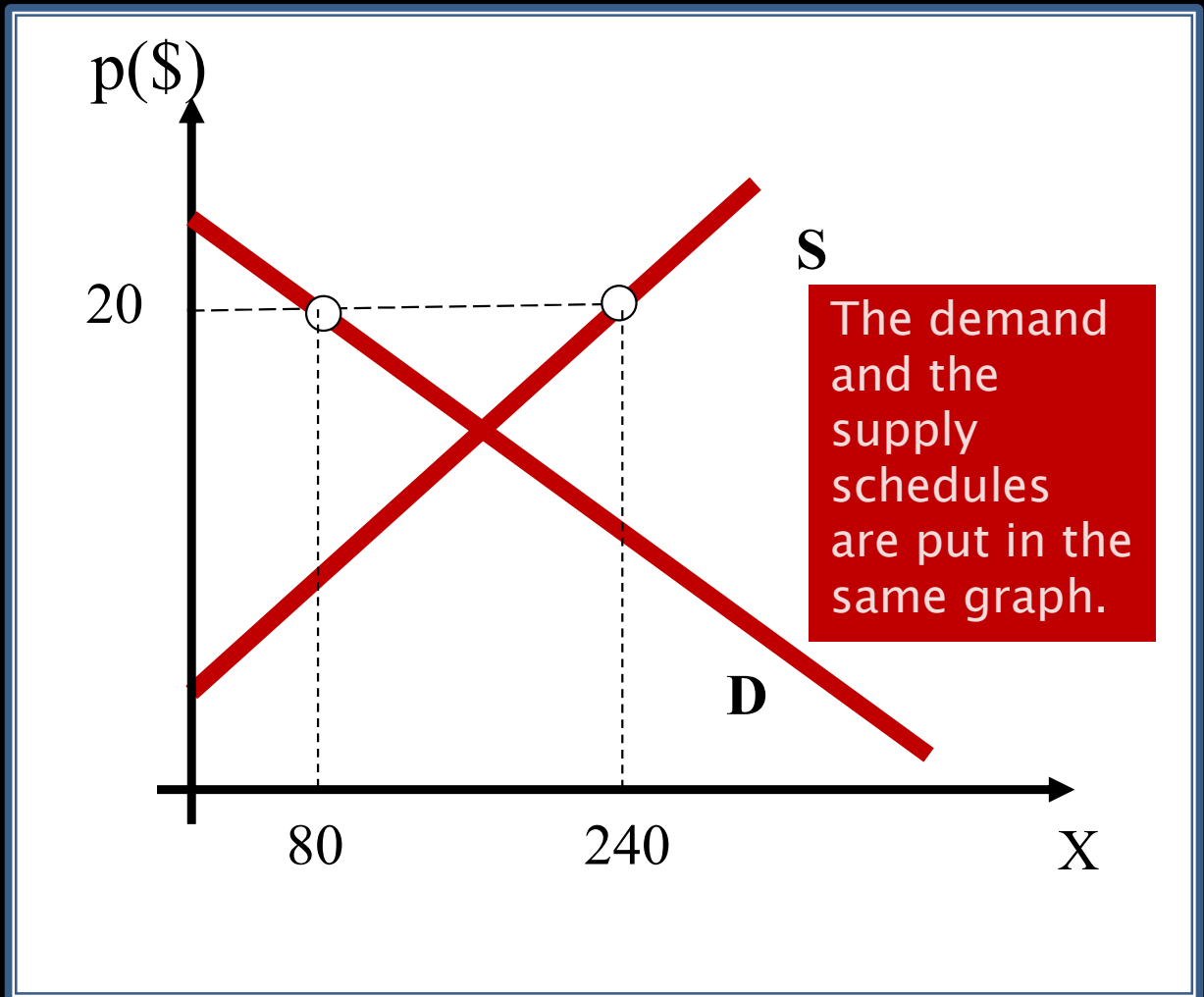
Understanding market forces

Excess supply

At price \$20 consumers demand 80 units, but producers want to supply 240 units.

So, if the price is at \$20, there will be 160 units unsold in the market.

Price must fall.

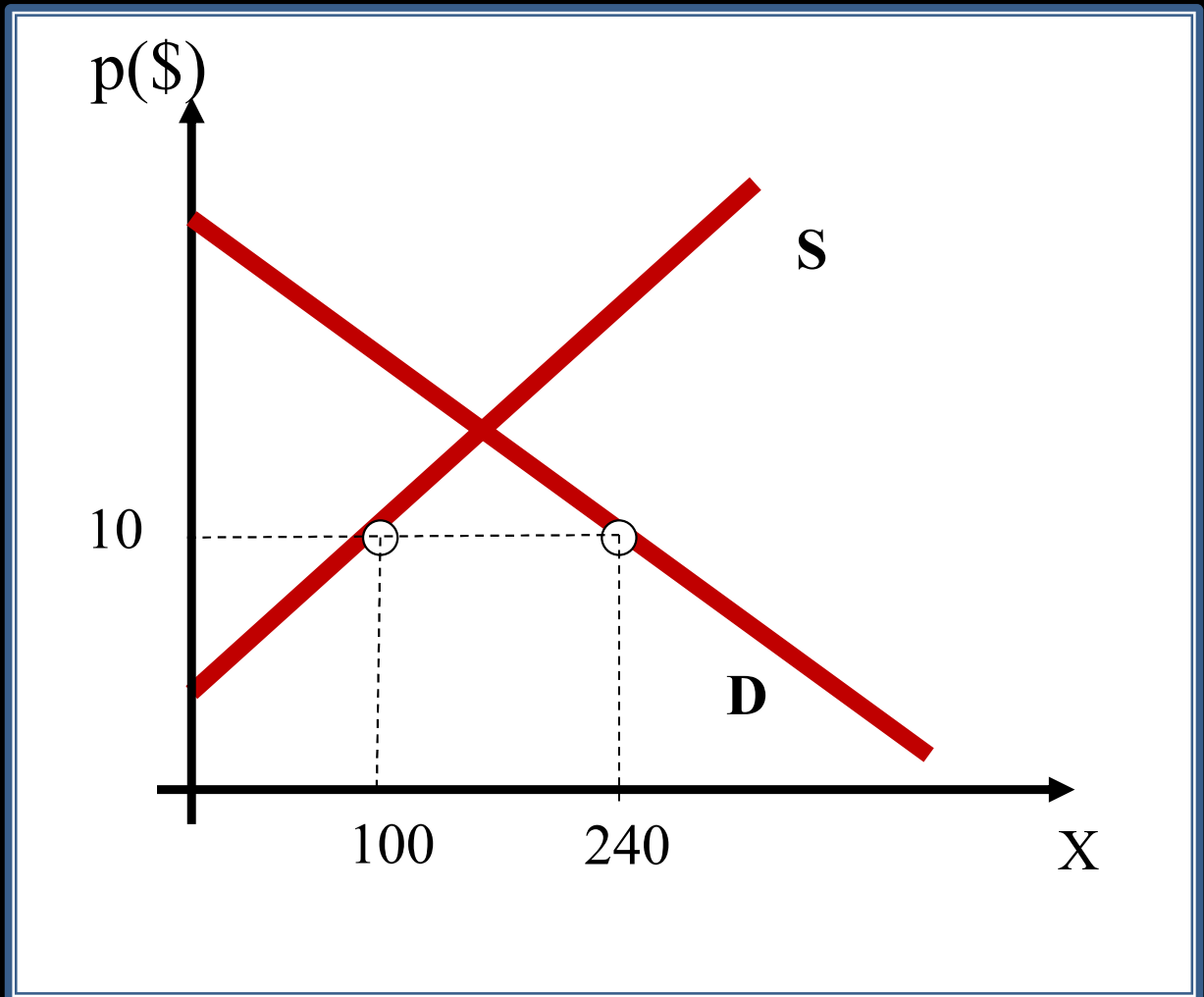


Excess demand

At price \$10 consumers demand 240 units, but producers want to supply 100 units.

So, if the price is at \$10, there will be excess demand of 140 units.

Price must rise.

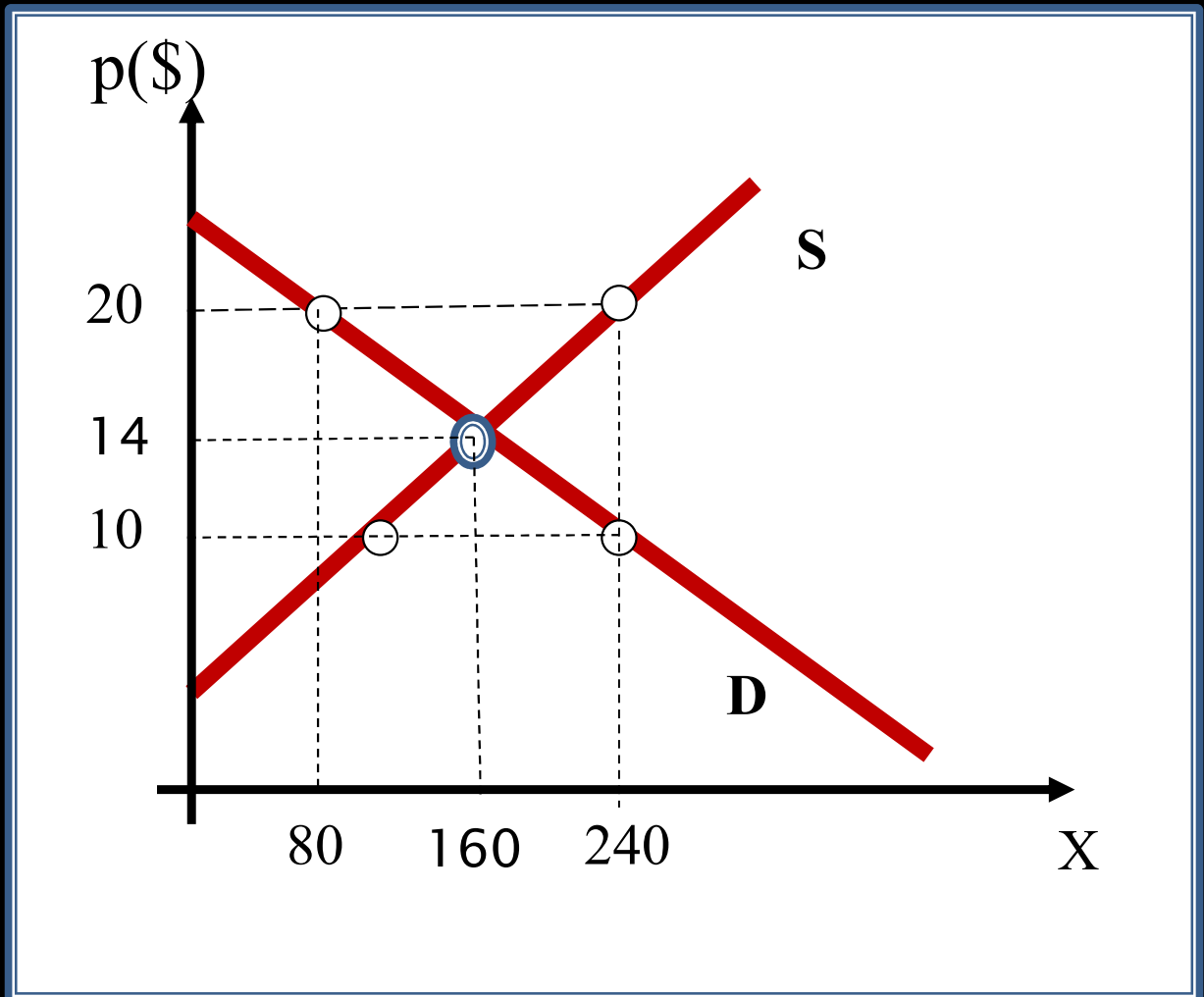


Market Equilibrium

Price must be something between \$10 and \$20.

At price \$14, the consumers demand 160 unit and the producers want to supply 160 units.

Market clears

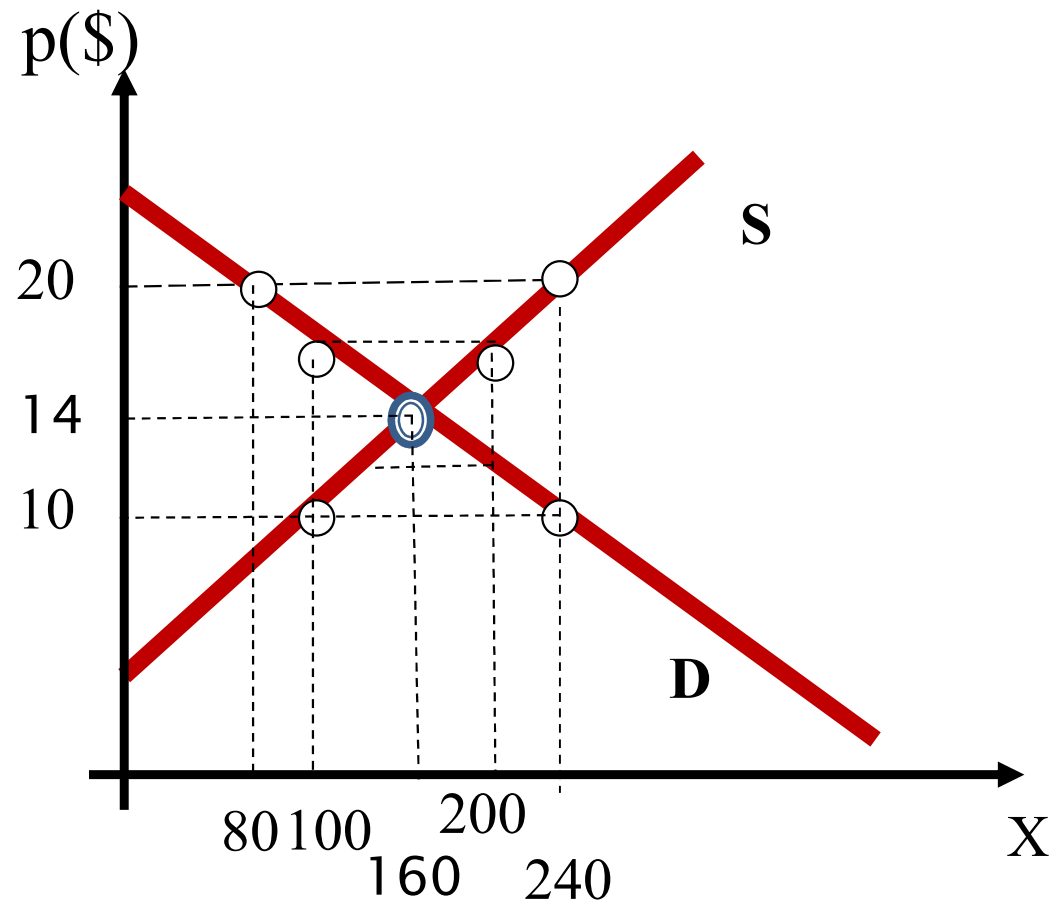


Price convergence to equilibrium

There is excess supply of 160 units at $p = 20$. To sell 240, price must be reduced to \$10. Market closes

At price \$10, there is excess demand of 140 units. Price rises. But does not go to 20.

Price fluctuates but converges to the equilibrium.



Price divergence (Different demand and supply schedules)

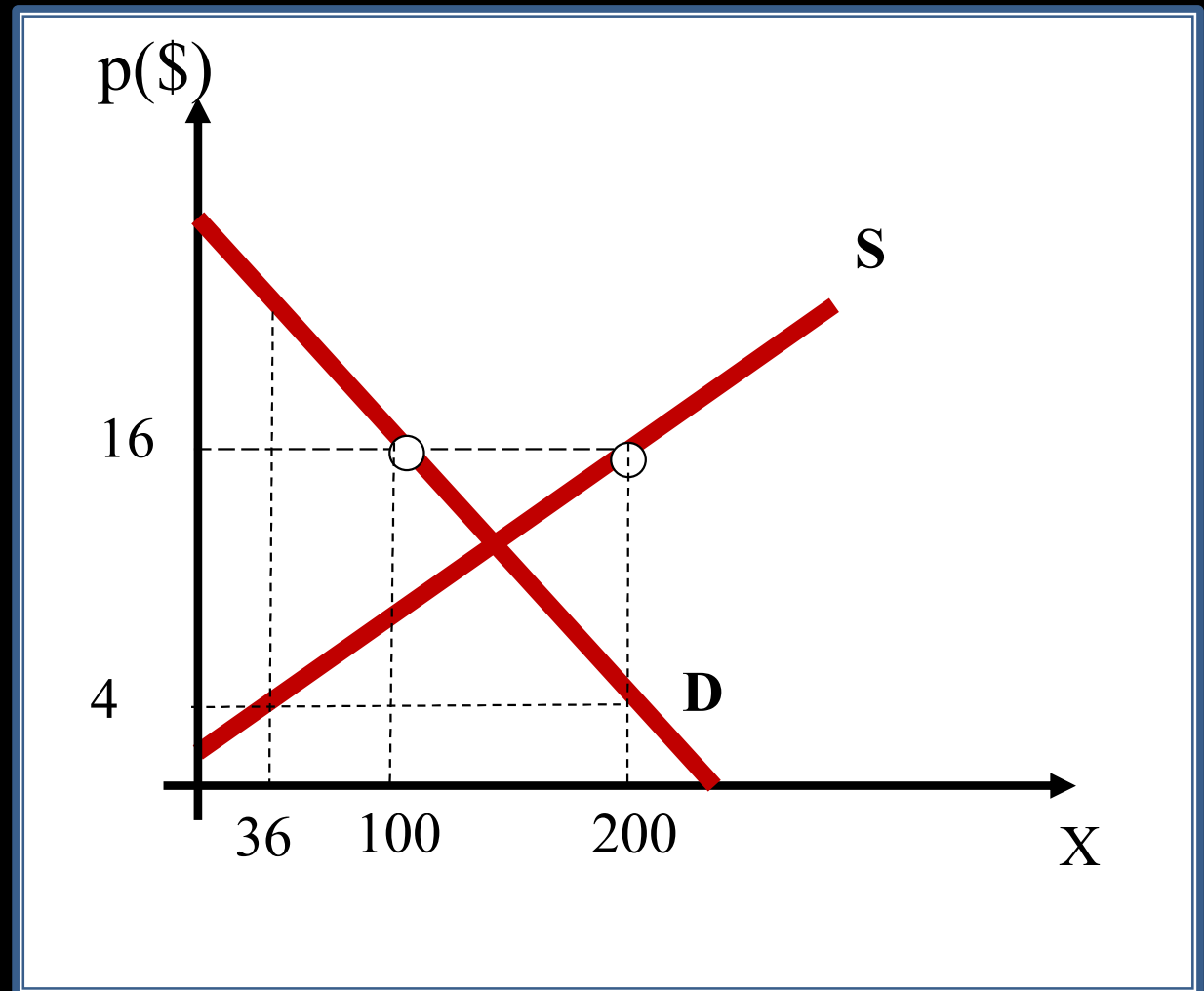
Suppose there is excess supply at $p = 16$.

To sell 200, price must be reduced to \$4.

Market closes

At price \$4, there is excess demand of 164 units. Price rises.

Price fluctuates and diverges away from the equilibrium.



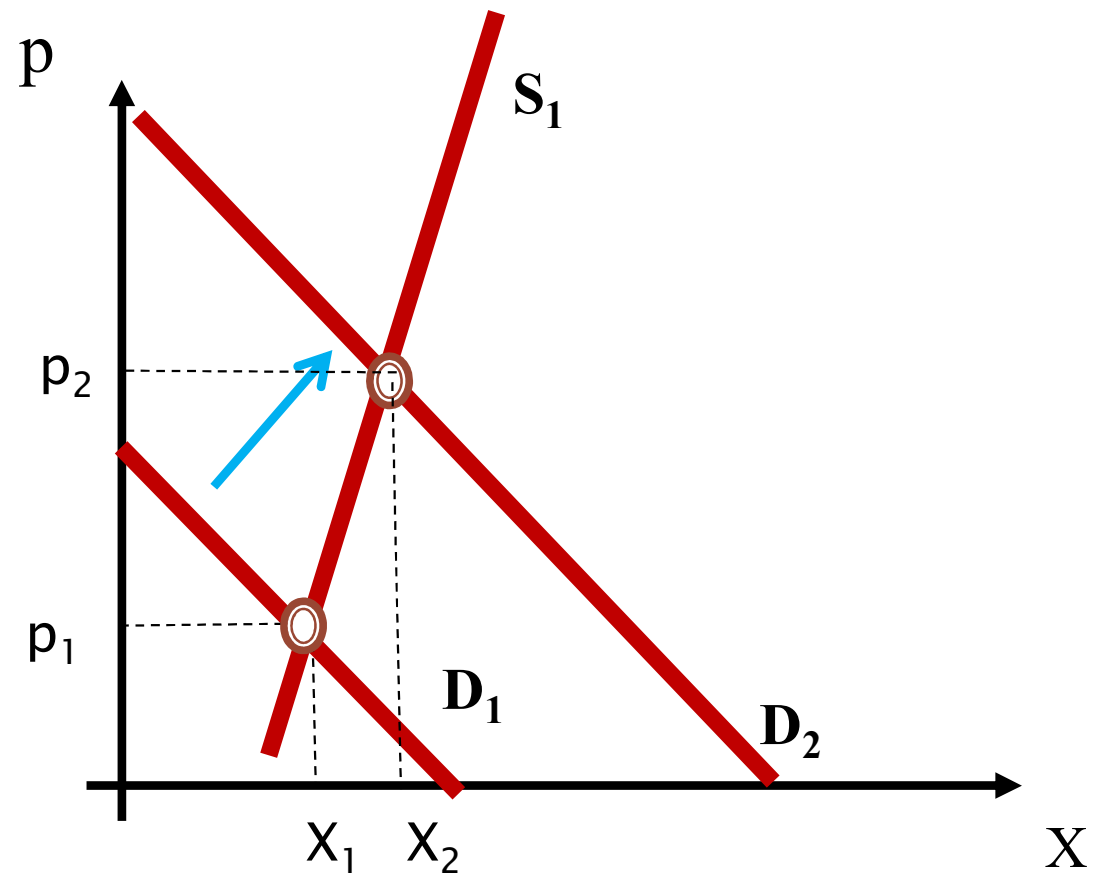
Case: Global Crude Oil Market

- ▶ Who demands?
 - The world
- ▶ Who supplies?
 - 80% of the supply comes from 12 countries.
- ▶ How is the price determined?
 - Through the forces of demand and supply.



Demand side effect on price

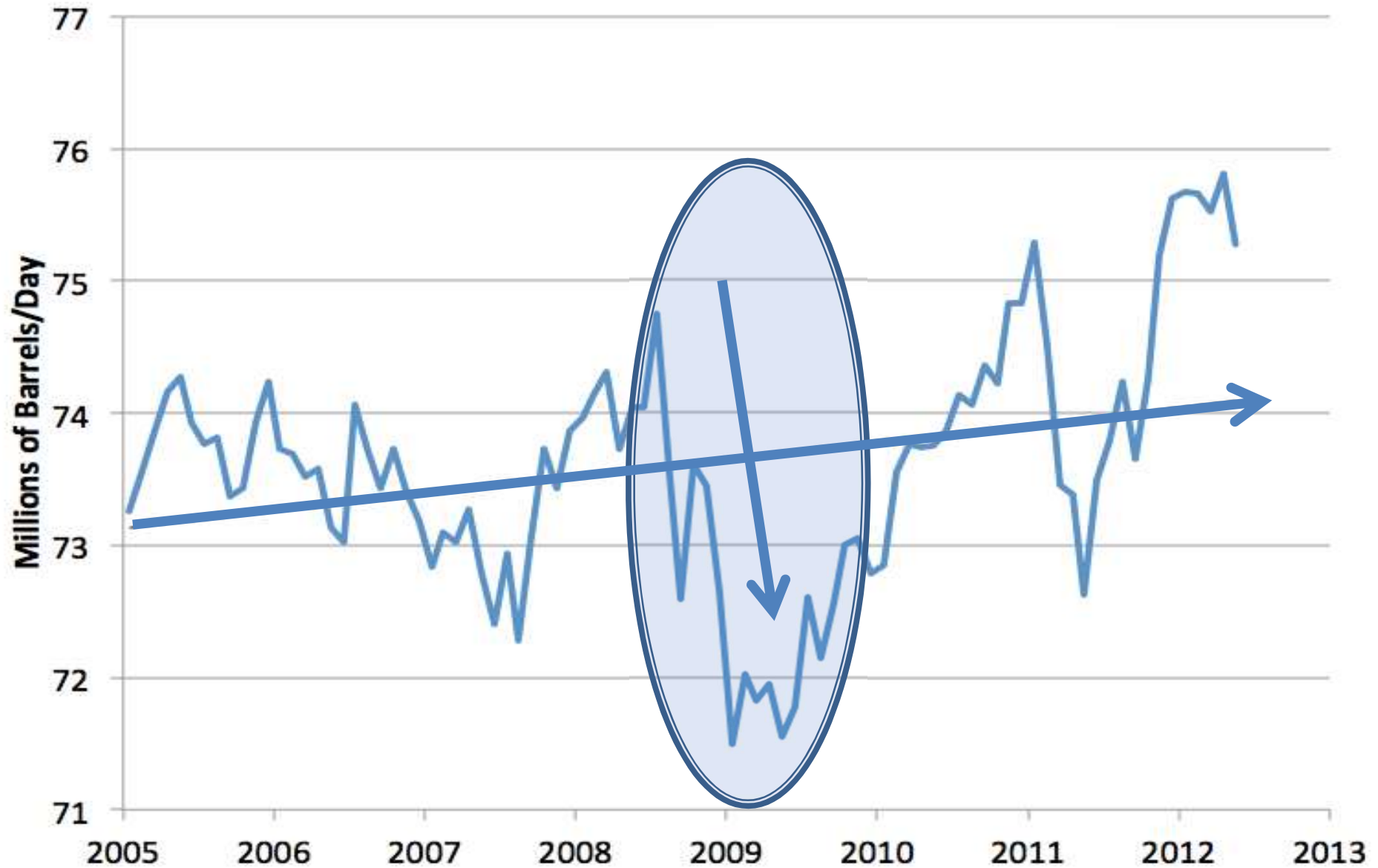
- ▶ Increase in demand shifts the demand schedule from D_1 to D_2
- ▶ Supply being more inelastic price increases sharply.
- ▶ Not much change in equilibrium output.
- ▶ This explains general increasing trend of oil price.



Demand shock – effect on price



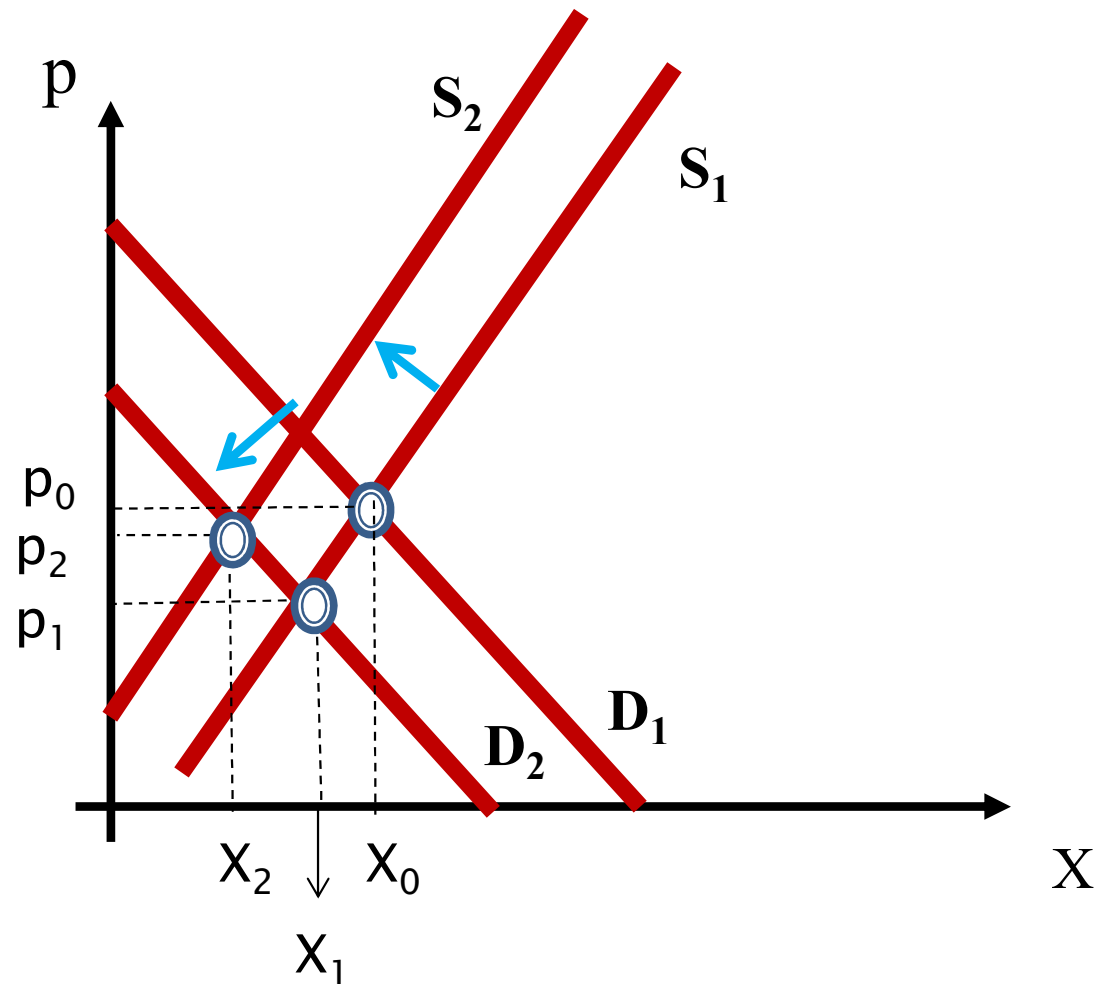
Supply response



Supply response to demand shock and effect on price

Initially price was p_0
Demand reduces to D_2
Price reduces to p_1 .
Equilibrium qty reduces to X_1
Supply gets adjusted

As supply is contracted price increases.



Oil price movement since the financial crisis



Why oil price dropped between 2014–16?

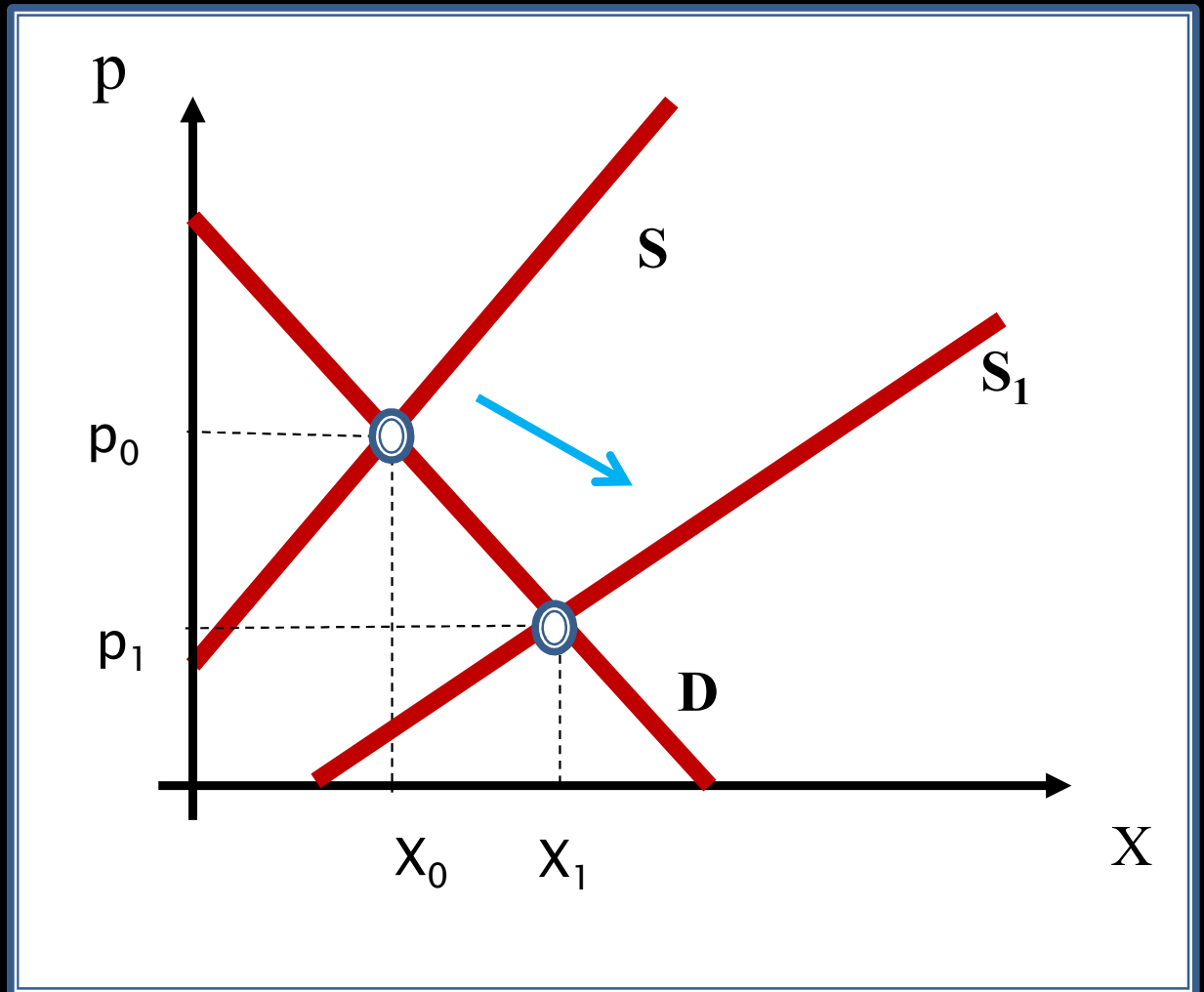
- Reduced demand?
 - No. Demand increased.
 - Average daily demand increased by approximately 2 million barrels per day in 2014 as compared to previous year.
- Supply side explanation
 - Supply increased.
 - US production increased steadily.
 - Fishbone fracking in North Dakota and Texas.
 - In general cost of fracking reduced.
 - OPEC could not reduce supply.

Increase in supply results in price fall

Supply schedule shifts right. The new supply schedule is S_1

Price falls from p_0 to p_1

X_1 quantity is demanded and sold.



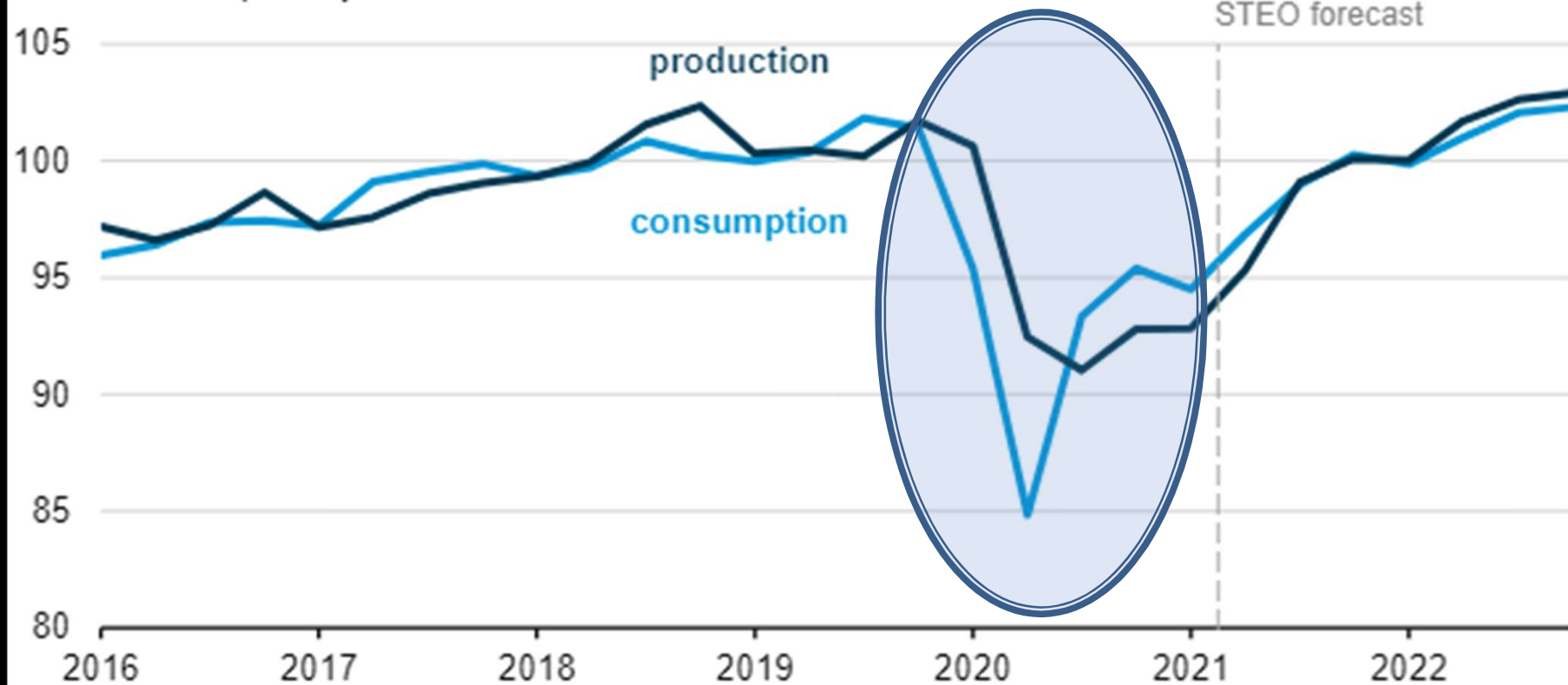
Oil price movement since the financial crisis



Supply-demand gap

Quarterly world liquid fuels production and consumption (2016–2022)

million barrels per day



Oil price movement during the Ukraine–Russia war

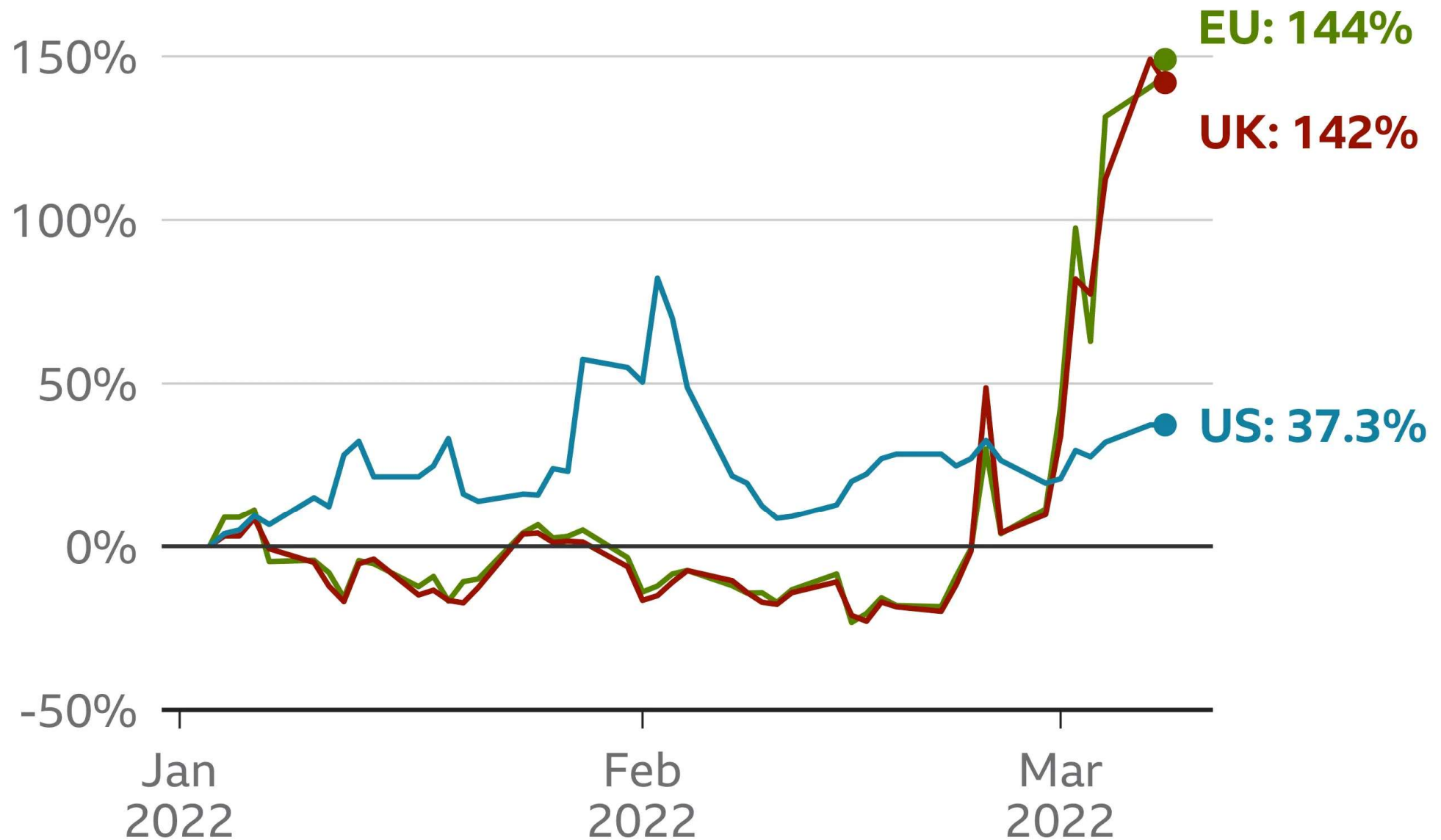
Oil price and important events



Note: Daily Brent prices and important events. Red lines show 2 mb/d cuts by OPEC+, 1.16 mb/d cuts by OPEC+; 1 mb/d cut (Saudi Arabia), 1.3 mb/d cut extension (Saudi Arabia and Russia), and 2.2 mb/d cut continuation of extensions by OPEC+. The yellow line shows the start of the conflict in the Middle East. The last observation is January 2, 2024.

Source: Bloomberg; World Bank.

Impact of input price on output price



Problem 1 – Set 1

Suppose that the market supply and demand curves for wheat are, respectively,

$$p = 0.002 X_s$$
$$p = 30 - 0.001 X_d$$

Prices are in ₹ per KG, and quantities in KG.

Government decided to buy enough wheat retain the price at ₹2 higher than the equilibrium price. The wheat bought by the government will be used in the National Rural Employment Program, which is outside the market system.

How much wheat (in KG) will the government buy?

Solution to Problem 1

- ▶ In eqm: $X_s = X_d \Rightarrow \frac{p}{0.002} = \frac{30-p}{0.001} \Rightarrow p = 20$
- ▶ Govt. sets $p = 20 + 2 = 22$
- ▶ At price of 22, $X_d = \frac{30-2}{0.001} = 8000$ and $X_s = \frac{22}{0.002} = 11000$
- ▶ Govt. buys the difference, i.e., 3000

Problem 2 – Set 1

Suppose that the government decided NOT to buy any wheat. Instead, they offer a subsidy per KG of wheat sold. How much subsidy (per KG) is required to retain the price at ₹2 higher than the equilibrium price derived in Problem 1?

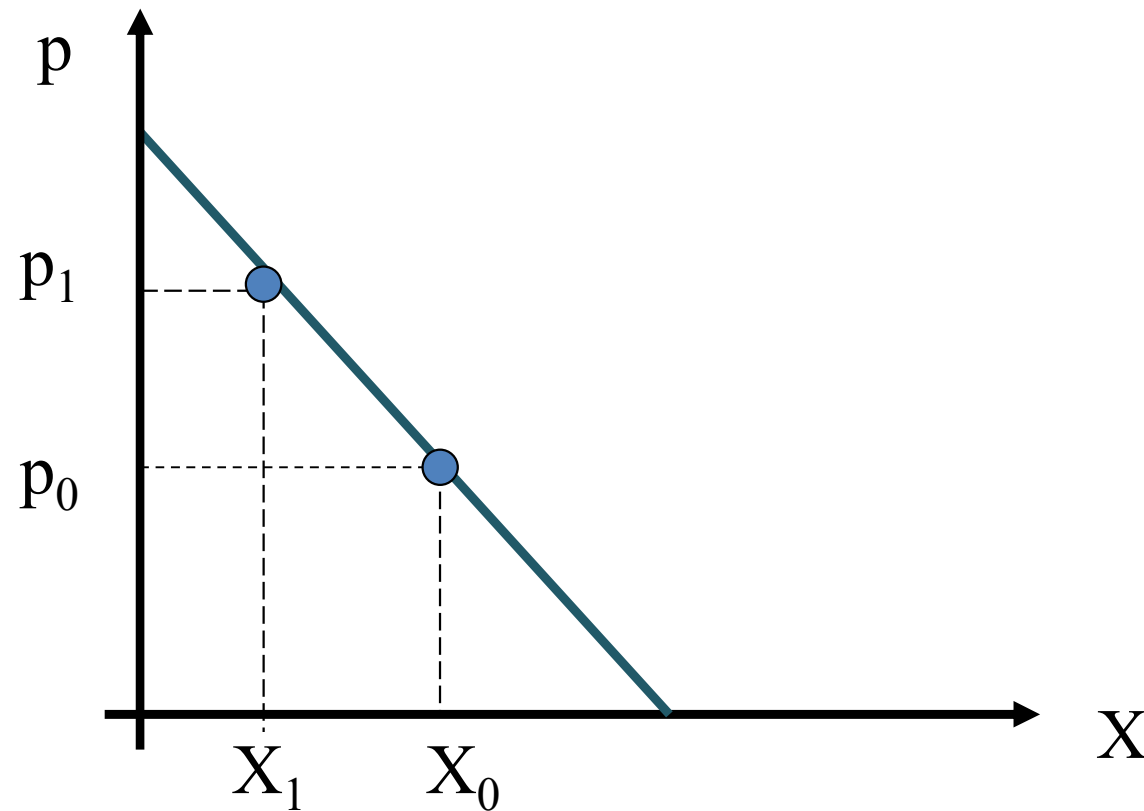
Which policy is more expensive for the government, the one given in Problem 1 or this one?

Solution to Problem 2

- ▶ Farmers get (inclusive of subsidy), $p' = 20 + 2 = 22$
- ▶ At $p' = 22$, how much is the supply?
- ▶ $X_s = \frac{22}{0.002} = 11000$
- ▶ At what price will 11000 KGs be demanded?
- ▶ Let it be p''
- ▶ Using the demand function: $p = 30 - (0.001 * 11000)$
- ▶ 11000 KGs will be demanded at $p'' = 19$
- ▶ Subsidy required $= 22 - 19 = 3$
- ▶ Govt. exp $= 11000 * 3 = ₹ 33,000$
- ▶ Govt. exp in Prob 1 $= 3000 * 22 = ₹ 66,000$

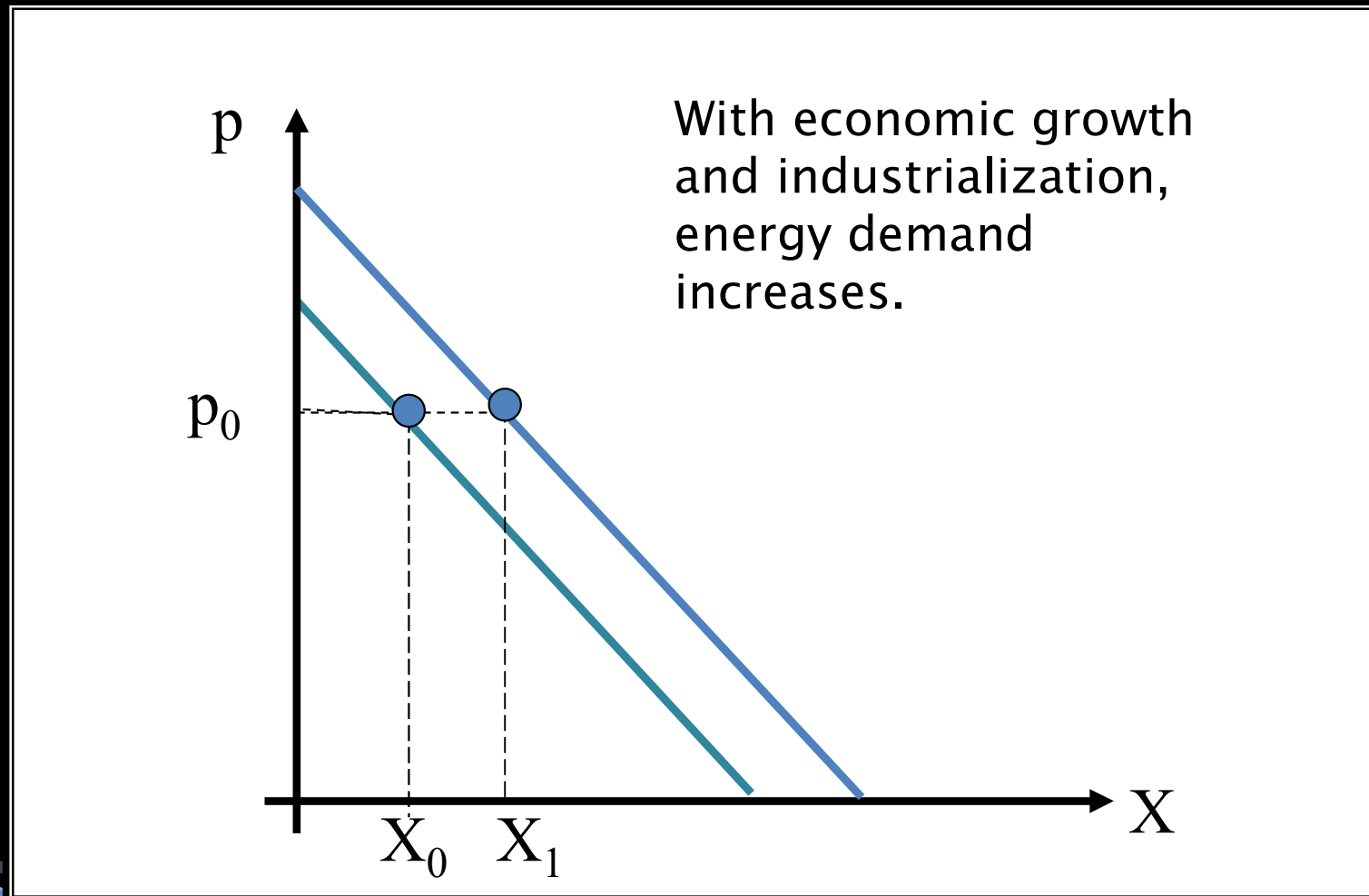
Understanding market demand

Change in quantity demanded for a change in price of the product (other factors remaining constant)



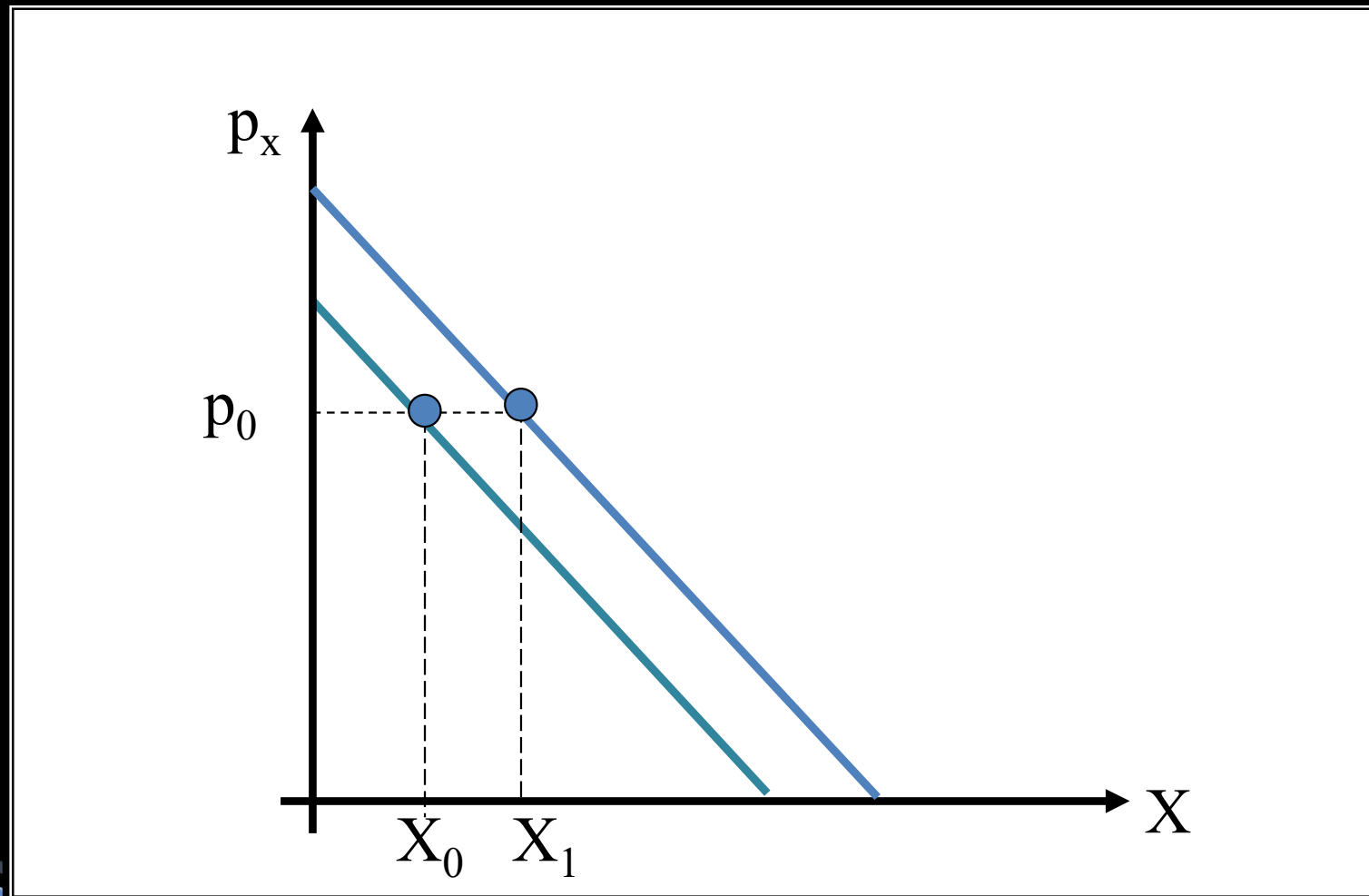
Shift of the demand curve – change in demand

For all prices demand increases (or decreases) due to change(s) in exogenous factor(s)



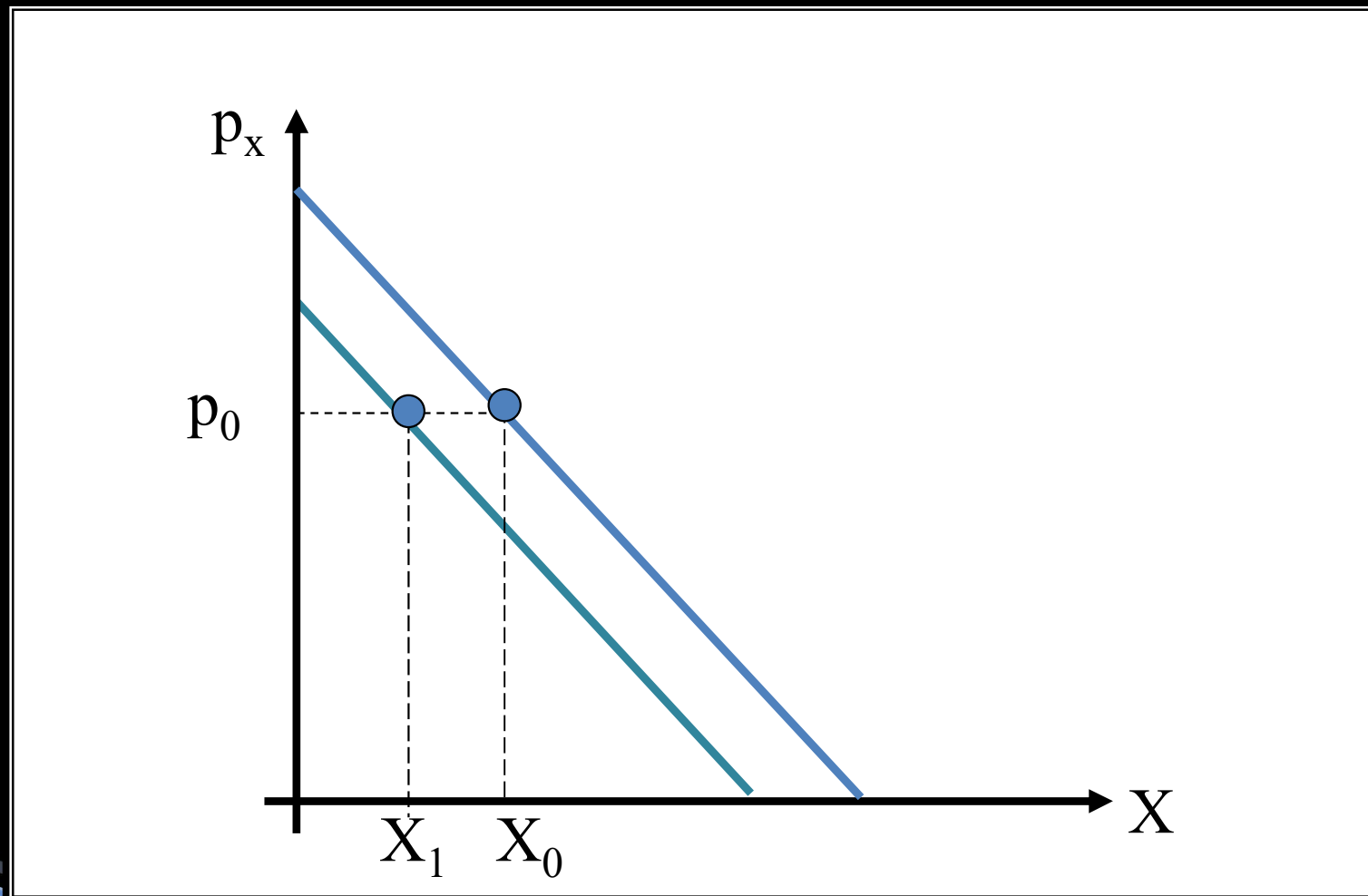
Shift of the demand curve – change in price of substitutes (or availability of new substitutes)

Demand for gasoline is expected to reduce with the adoption of electric vehicles (EVs)



Shift of the demand curve – change in price of complements

With an increase in the price of gasoline vehicles (GVs), demand for gasoline reduces.



Managerial Economics

Impact of demand on revenue and profit
Elasticity of demand
Individual Choices and Decisions
Consumers' behaviour –

Term I

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Importance of market demand for revenue and profit

Definitions: Revenue, cost and profit

$$\text{Revenue}[R] = \text{Quantity Sold } [x] \cdot \text{Price } [p]$$

$$\text{Cost } [C] = \text{Var. Cost } [c(x)] + \text{Fixed Cost } [F]$$

Will be discussed later. For now, $C = c.x$, where c = unit cost

$$\begin{aligned}\text{Profit } [\pi] &= \text{Revenue}[R] - \text{Cost } [C] \\ &= p.x - c.x = D(x)x - c.x\end{aligned}$$

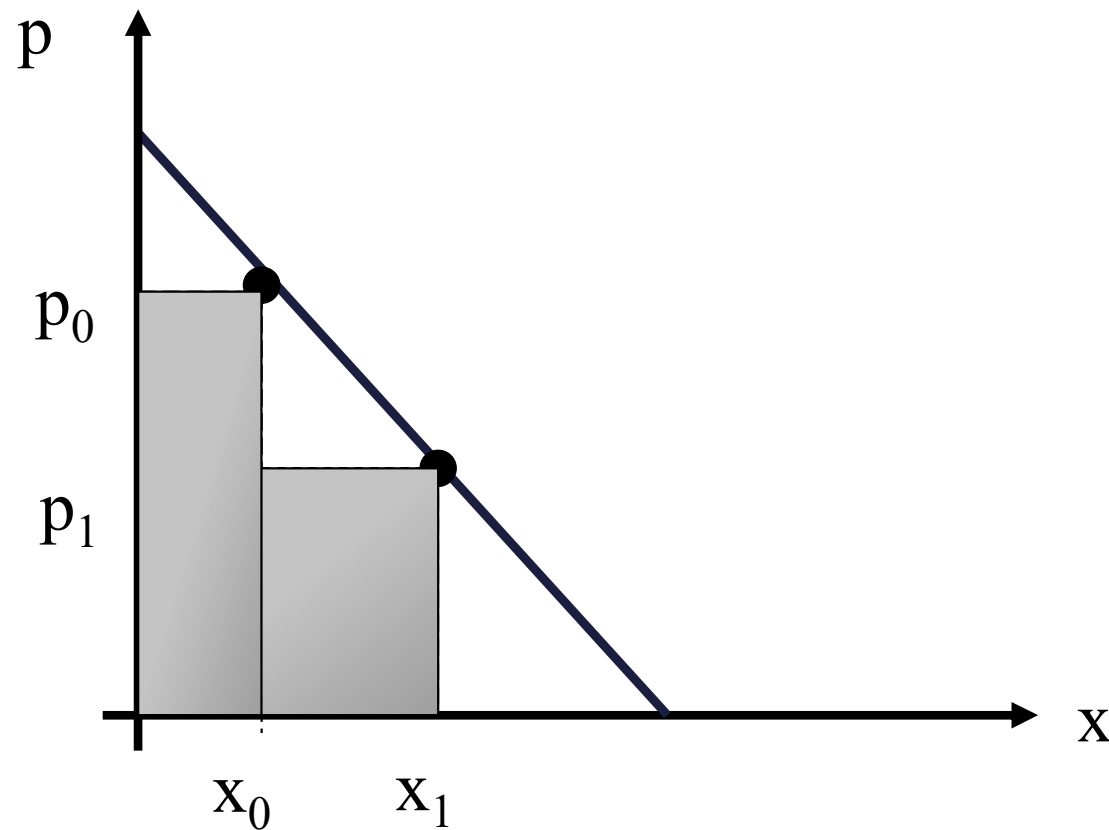


Demand function: $p = D(x)$

Revenue and the demand function

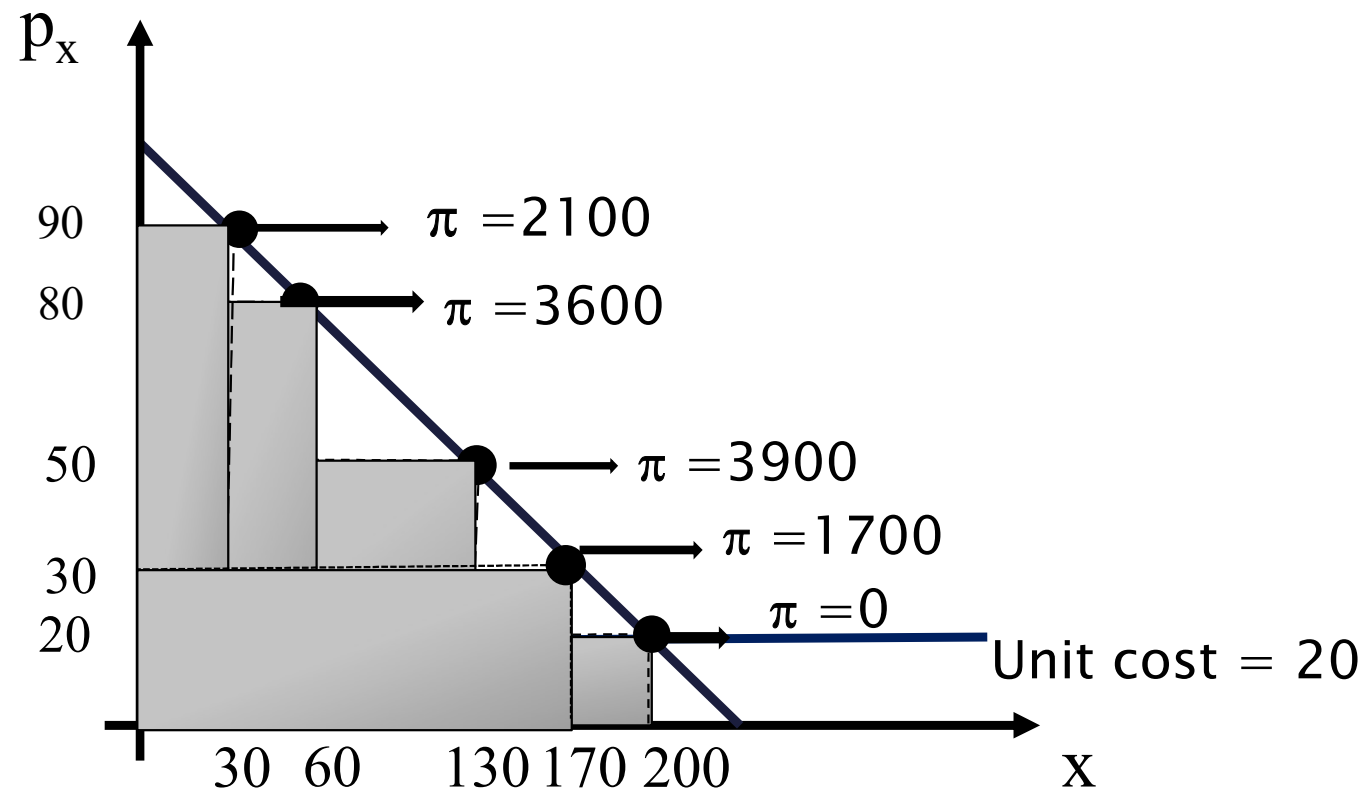
Revenue(R) = Quantity Sold (x) . Price (p)

$$R = xD(x)$$



Profit and the demand function

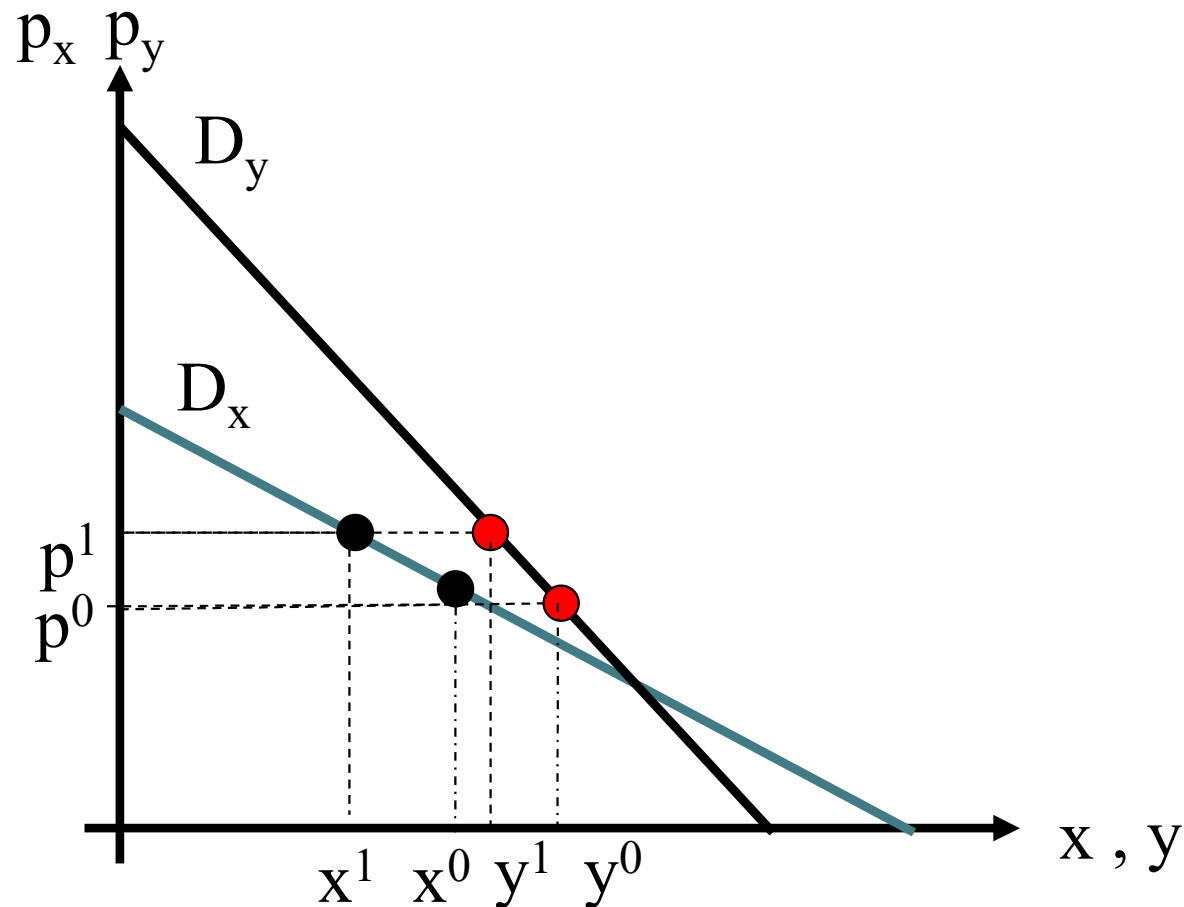
What should be your price?



Importance of demand elasticities

How are these demand curves different?

X and Y are two products. Are the consumers of X more price sensitive or the consumers of Y?



Price elasticity of demand – a measure of price sensitivity of consumers

- Price elasticity of demand captures the responsiveness of quantity demanded to changes in price

$\varepsilon = (\text{percentage change in quantity demanded}) / (\text{percentage change in price})$

$$= \left(\frac{dx}{dD(x)} \right) \frac{D(x)}{x}$$
$$= \frac{p_x dx}{x d_x}$$

$$\varepsilon < 0 \text{ as } \frac{dx}{dD(x)} < 0 \quad [\text{Demand fn is downward sloping}]$$

- Suppose, other things remaining constant, an increase in your price from Rs. 10 / unit to Rs. 12 / unit resulted in sales drop from 15000 units / day to 13000 units / day. Can you calculate the price elasticity of demand for your product?

Utility function - Definition

A one to one correspondence between quantity of a 'good' consumed and utility derived from that consumption.

- ▶ The utility function of a typical consumer
 - increases with increase in consumption till the point of satiation (**increasing** function)
 - increases at a decreasing rate (**concave** function)
 - for every marginal (infinitesimally small) increase in consumption there is increase in utility (**continuous** function)

Marginal Utility - Definition

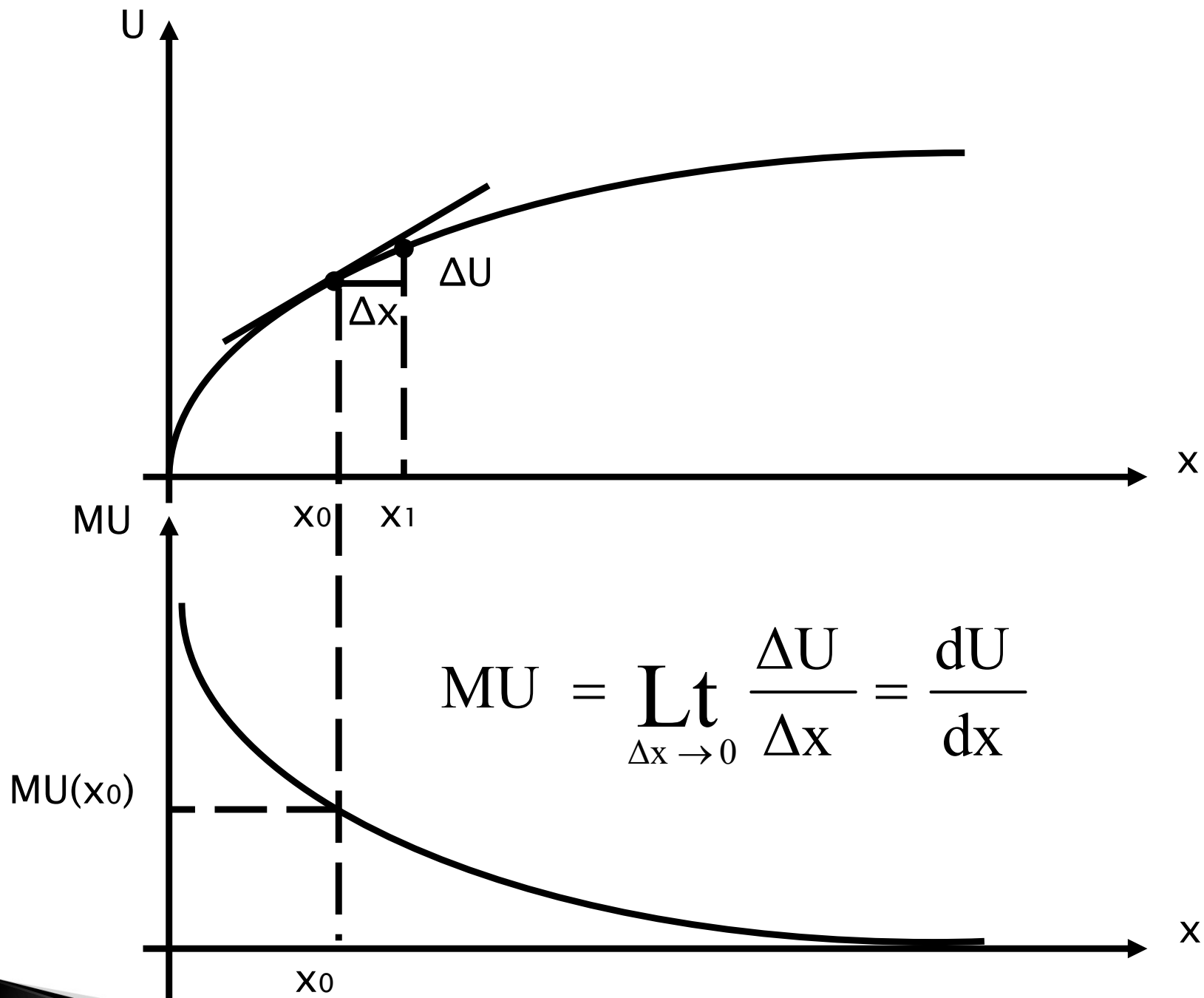
- ▶ Additional utility derived from consumption of an additional unit.

- ▶ (Technical definition)

Utility derived from marginal consumption.

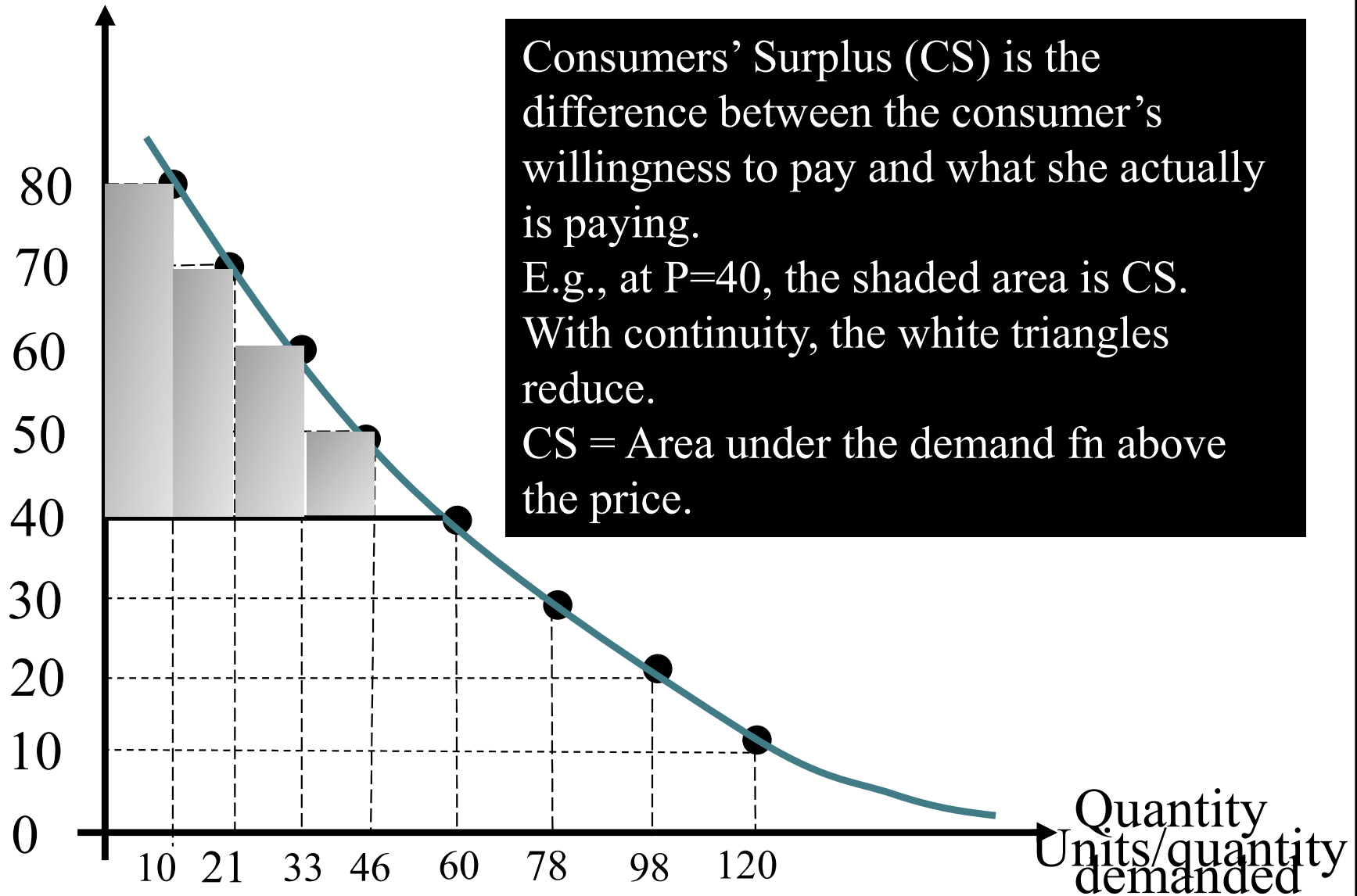
If $U = u(x)$

$MU = du(x)/dx$



Maximum WtP function = Demand function

Maximum WtP



Consumers' psychology

- ▶ More is better
 - Utility increases with consumption, i.e., $MU > 0$
- ▶ Diminishing Marginal Utility:
 - Marginal Utility decreases with increase in consumption.

Two-goods framework

- ▶ Consumer's optimization problem:

$$\text{Max } U(x, y)$$

$$\text{S.t. } p_x x + p_y y \leq M$$

The laws of More is Better and DMU holds for each good in this two-good (two variables) utility function.

$\frac{\partial U}{\partial x}$ and $\frac{\partial U}{\partial y}$ are both non - negative and decreasing

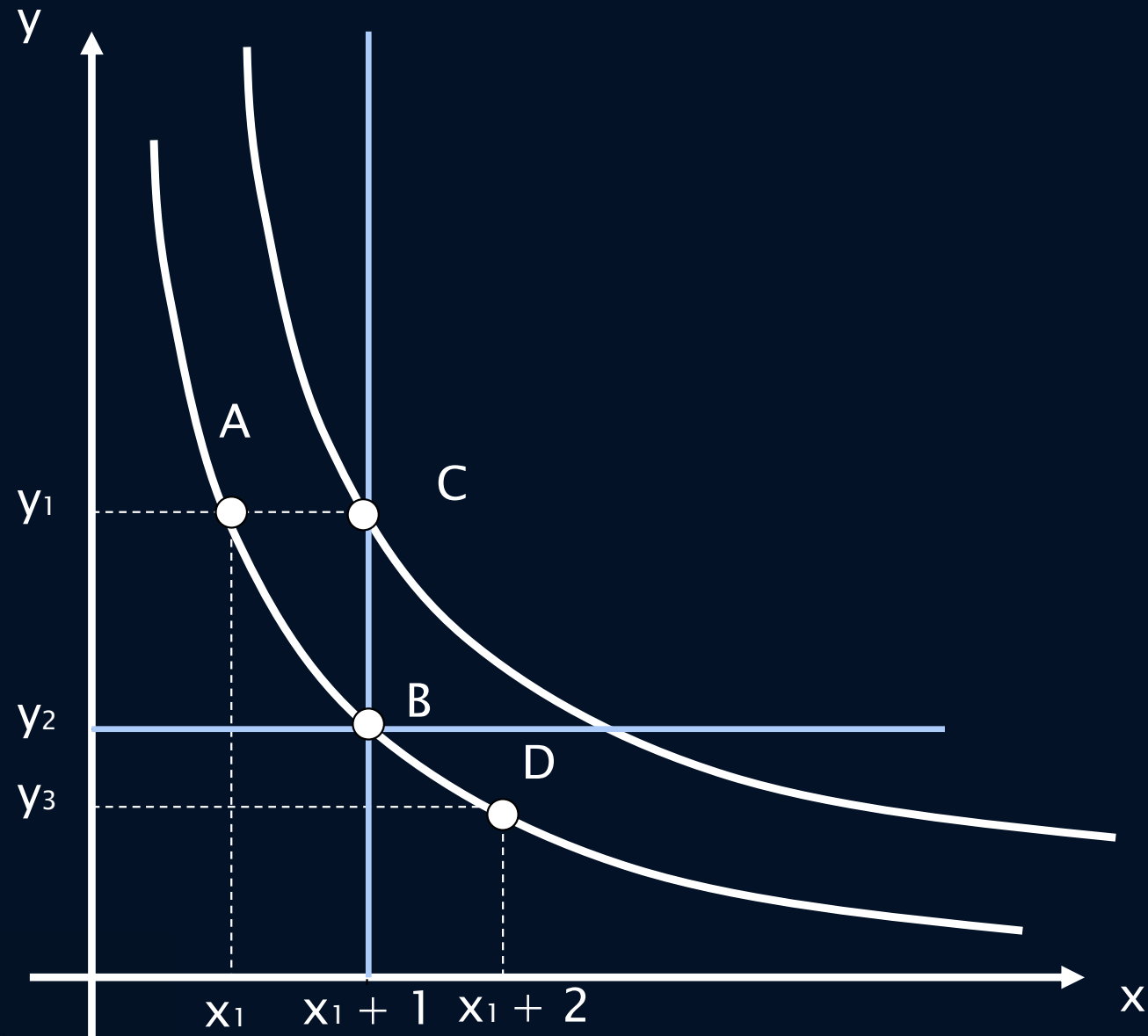
Indifference curve

- Locus of points on the consumption space that generates the same utility.

The consumption space is the non-negative Cartesian space where quantities of different commodities are measured on different axes.

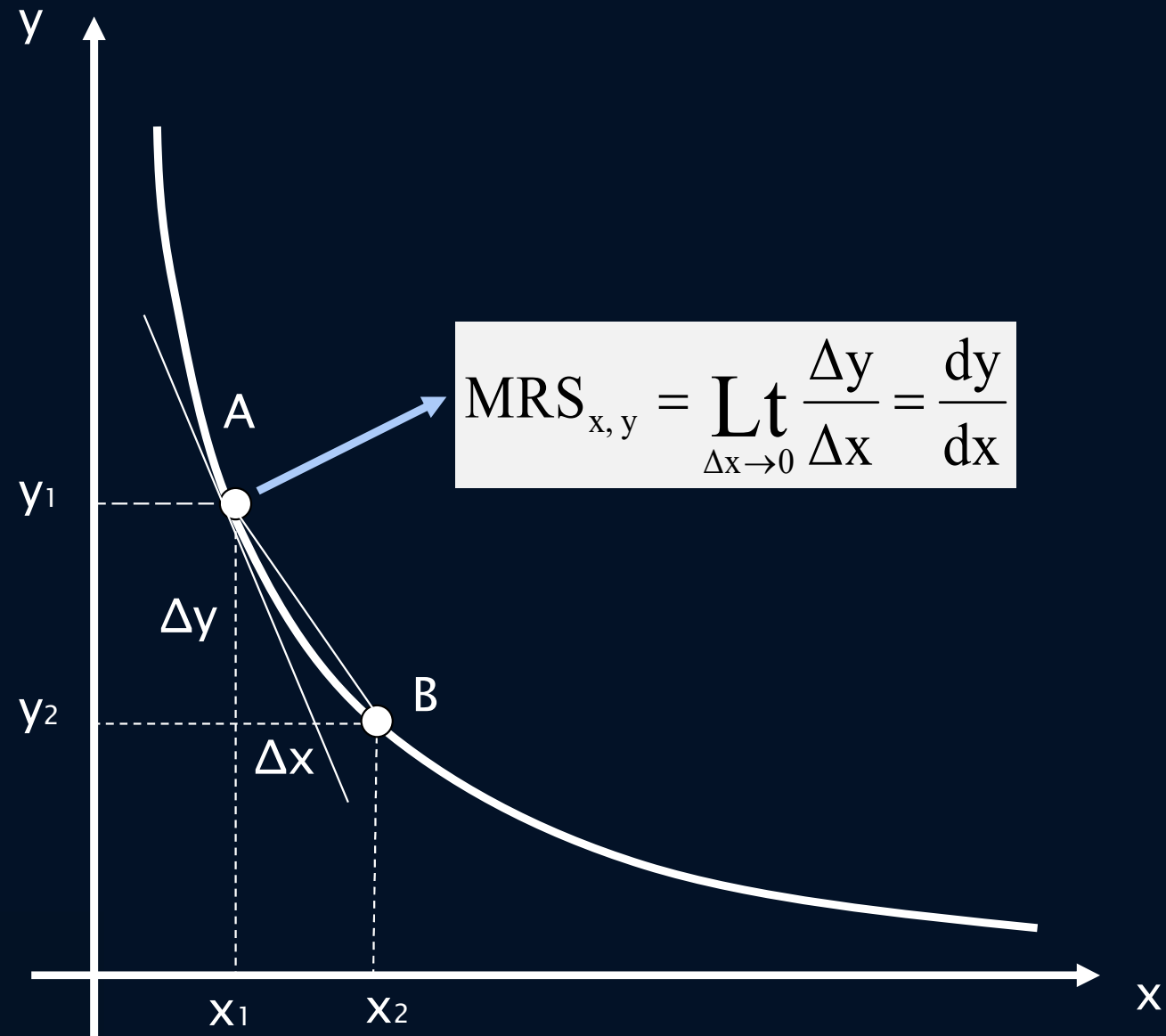
- **Properties:**
 1. Downward sloping
 2. Convex to the origin

Indifference curve slopes downward



Marginal rate of substitution

Amount of Y the consumer is willing to give up to get one extra unit of X, such that her utility remains unchanged



Problem 3 – Set 1

1. A consumer's utility function is given by $U = x^{1/3}y^{2/3}$.
 - Is the utility increasing in x ? Explain
 - Is the utility increasing in y ? Explain
 - Find the marginal utility of X as a function of x and y .
 - Find the marginal utility of Y as a function of x and y .
 - Is the MU of X decreasing? Explain
 - Is the MU of Y decreasing? Explain
 - Find the MRS when the consumer consumes 100 units of X and 200 units of Y

Solution to Problem 3 – Set 1

1. $U = x^{1/3}y^{2/3}$.

- $MU_x = \frac{\partial U}{\partial x} = \frac{y^{2/3}}{3x^{2/3}} > 0$, and $\frac{\partial^2 U}{\partial x^2} = -\frac{2y^{2/3}}{9x^{5/3}} < 0$
- $MU_y = \frac{\partial U}{\partial y} = \frac{2x^{1/3}}{3y^{1/3}} > 0$, and $\frac{\partial^2 U}{\partial y^2} = -\frac{2x^{1/3}}{9y^{4/3}} < 0$
- U is increasing at a decreasing rate in both x and y
- $MRS_{x,y} = -\frac{MU_x}{MU_y} = -\frac{y}{2x} = -\frac{200}{2*100} = -1$

Budget constraint

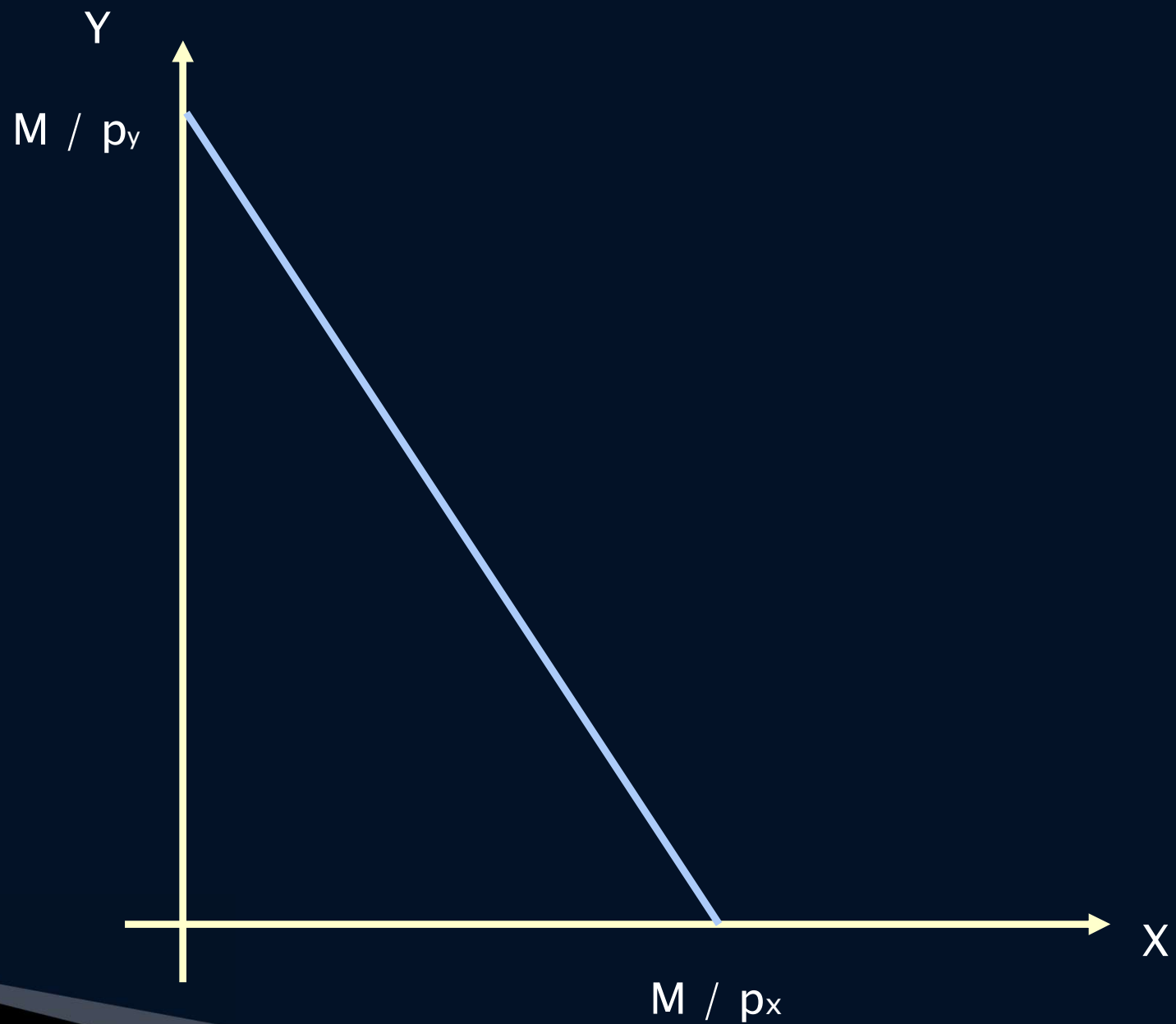
The budget constraint

Defines the feasible region of consumption on the consumption space, for a given price-income situation.

The Budget Equation

$$x \cdot p_x + y \cdot p_y = M$$

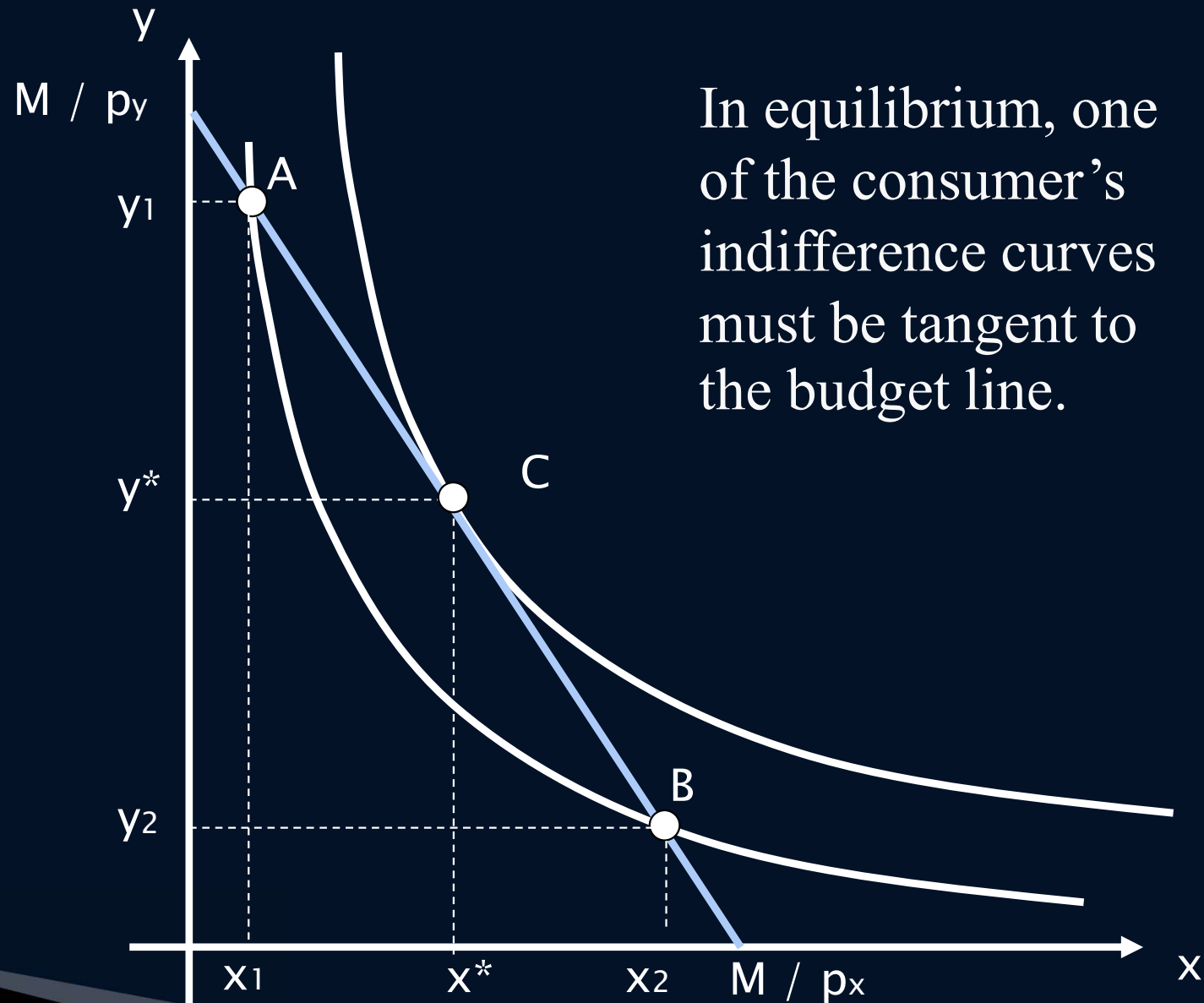
Budget line



Consumers' Equilibrium

- **The point on the consumption space where the consumer maximizes her utility, subject to her budget constraint.**
There is no incentive for the consumer to deviate from this point.

Consumers' Equilibrium - Graphically



Consumers' Equilibrium - Condition

The slope of the budget line = the slope of the indifference curve

=>

The price ratio = the absolute value of the marginal rate of substitution

$$\Rightarrow \frac{MU_x}{MU_y} = \frac{p_x}{p_y} \quad \Rightarrow \frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

Problem 4 – Set 1

A consumer's utility function is given by $U = x^{0.5}y^{0.5}$. What is her $MRS_{X \text{ for } Y}$ when she consumes 5 units of Y and 1 unit of X? Suppose, her income is 300 and the prices of X and Y are 2 and 1 respectively. Is she on her equilibrium? How much X and Y should she consume at her equilibrium?

Solution to Problem 4 – Set 1

1. $U = x^{0.5}y^{0.5}$.

- $MU_x = \frac{\partial U}{\partial x} = \frac{y^{0.5}}{2x^{0.5}} > 0$, and $\frac{\partial^2 U}{\partial x^2} = -\frac{y^{0.5}}{4x^{1.5}} < 0$

- $MU_y = \frac{\partial U}{\partial y} = \frac{x^{0.5}}{2y^{0.5}} > 0$, and $\frac{\partial^2 U}{\partial y^2} = -\frac{x^{0.5}}{4y^{1.5}} < 0$

- $MRS_{x,y} = -\frac{MU_x}{MU_y} = -\frac{y}{x} = -\frac{5}{1} = -5$

- Budget eq: $2x + y = 300$

- Slope of BL. $-\frac{p_x}{p_y} = -\frac{2}{1} = -2$

- Not in equilibrium as, $MRS_{x,y} \neq$ slope of BL

Solution to Problem 4 contd.

1. In the equilibrium, $MRS_{x,y} = -\frac{p_x}{p_y}$, or, $\frac{y}{x} = \frac{p_x}{p_y}$
2. Or, $2x = y$ [Since, $p_x = 2$ and $p_y = 1$]

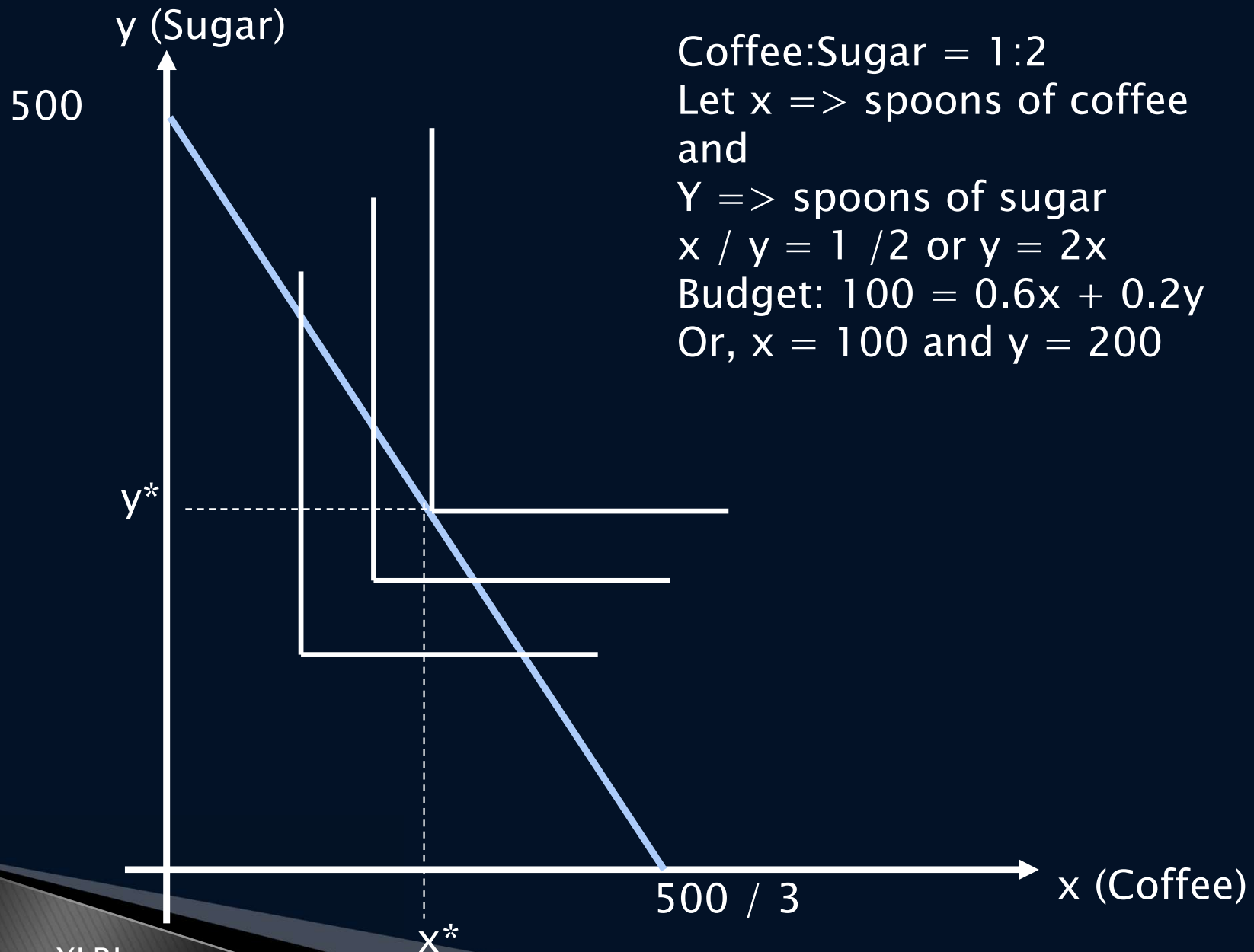
Equilibrium condition

- Since the equilibrium point is on the BL, the condition must satisfy the Budget eq: $2x + y = 300$
- Substituting we get, $2y = 300$, or $y = 150$. Hence, $x = 75$.
- Check, the budget eq. is satisfied.

Problem 5 – Set 1

Dr. Wooster consumes two goods – coffee and sugar. He consumes exactly two teaspoons of sugar per teaspoon of coffee. Graph Wooster's indifference curves in the coffee – sugar space. If Wooster has \$100 to spend on coffee and sugar, and sugar costs \$0.20 per teaspoon and coffee costs \$0.60 per teaspoon, how much coffee and sugar does he buy?

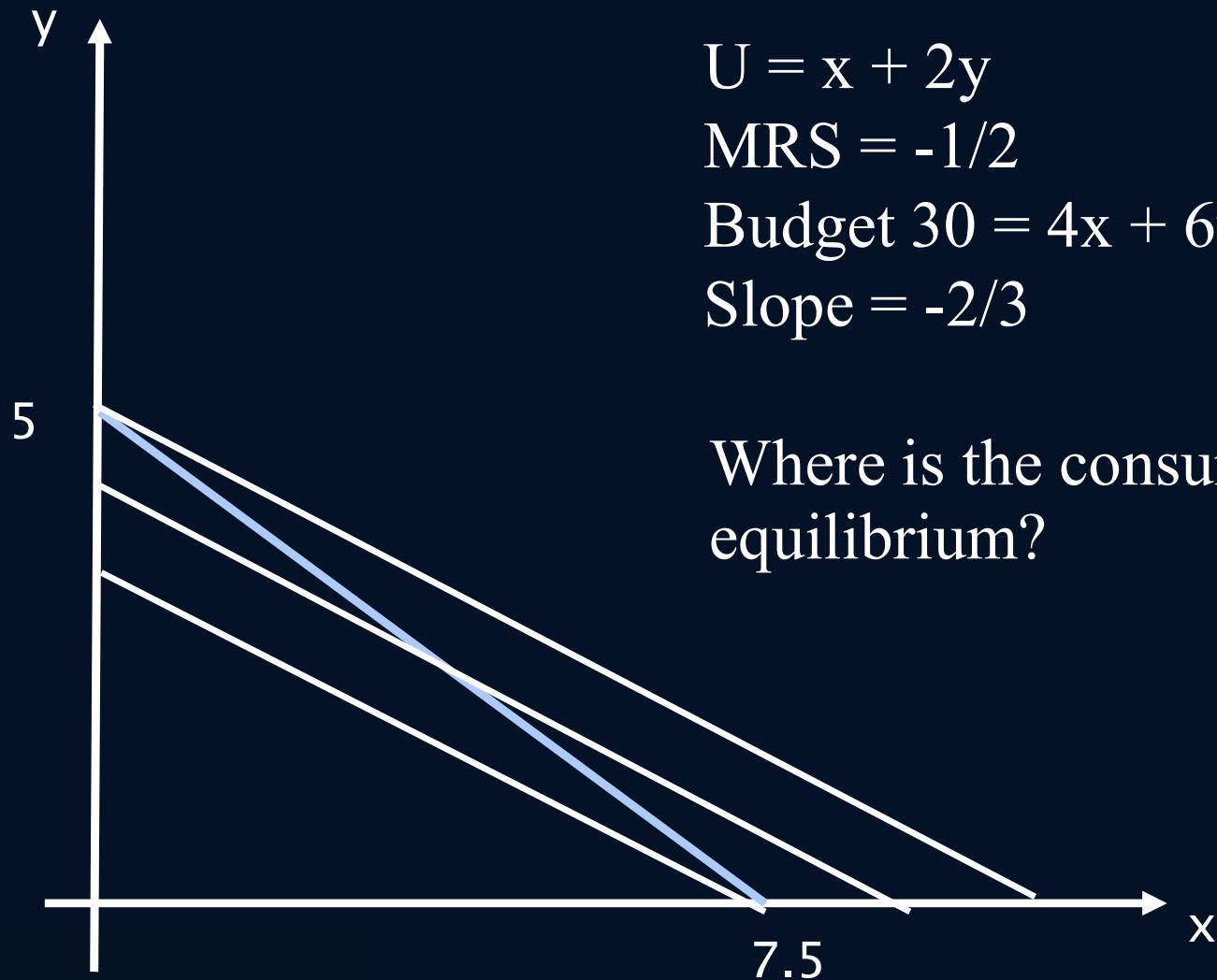
Solution to Problem 5 – Set 1 (Perfect complements)



Problem 6 – Set 1

A consumer's utility function is given by $U = x + 2y$. Draw the consumer's indifference map. What is her marginal rate of substitution? Suppose her budget is 30, price of x is 4, and the price of y is 6. How much x and y does she consume in the equilibrium?

Solution to Problem 6 – Set 1 (Perfect substitutes)



$$U = x + 2y$$

$$MRS = -1/2$$

$$\text{Budget } 30 = 4x + 6y$$

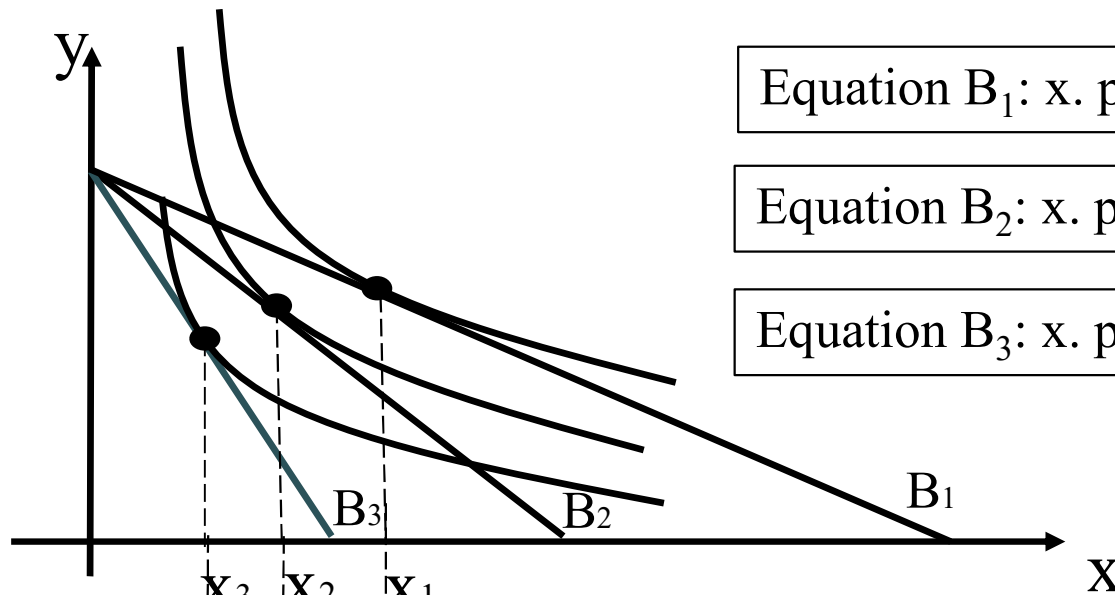
$$\text{Slope} = -2/3$$

Where is the consumers' equilibrium?

Individual Demand Function

- In the 2-goods world, consumers' equilibrium depends on three parameters: p_x , p_y and M . It is intuitive that the demand of a good for an individual is a function of these three variables.
- $x = D_x(p_x, p_y, M)$
- p_x is the endogenous variable and p_y and M are exogenous variables.
- In the general N-goods model, consumers' equilibrium depends on N prices $p_1, p_2, \dots, p_n$ and M . The consumer's demand for good i is:
 - $x_i = D_i(p_1, p_2, \dots, p_i, \dots, p_n, M)$

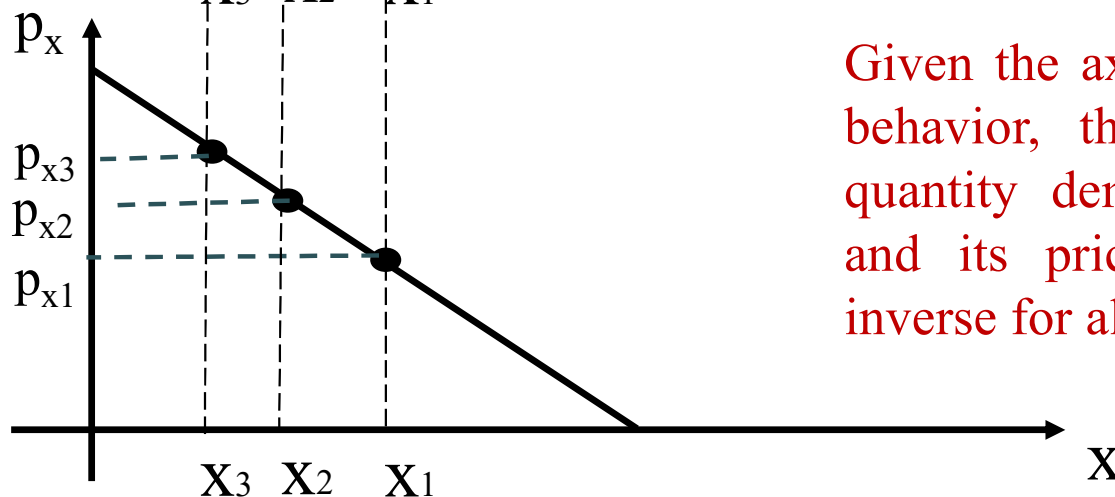
Consumer's equilibrium to consumer's demand



$$\text{Equation } B_1: x \cdot p_{x1} + y \cdot p_y = M$$

$$\text{Equation } B_2: x \cdot p_{x2} + y \cdot p_y = M \quad p_{x2} > p_{x1}$$

$$\text{Equation } B_3: x \cdot p_{x3} + y \cdot p_y = M \quad p_{x3} > p_{x2}$$



Given the axioms of consumers' behavior, the relation between quantity demanded for a good and its price is bound to be inverse for all non-Giffen goods.

Managerial Economics

Individual demand to market demand
Substitutes and complements

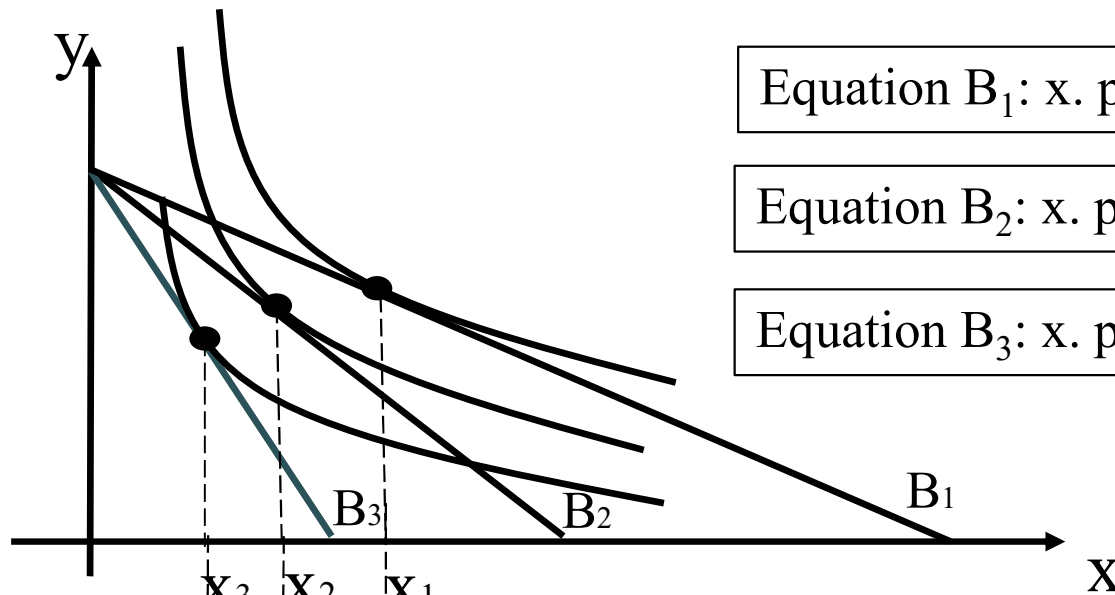
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Individual Demand Function

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- In the general N-goods model, consumers' equilibrium depends on N prices $p_1, p_2, \dots, p_n$ and M . The consumer's demand for good i is:
 - $x_i = D_i(p_1, p_2, \dots, p_i, \dots, p_n, M)$

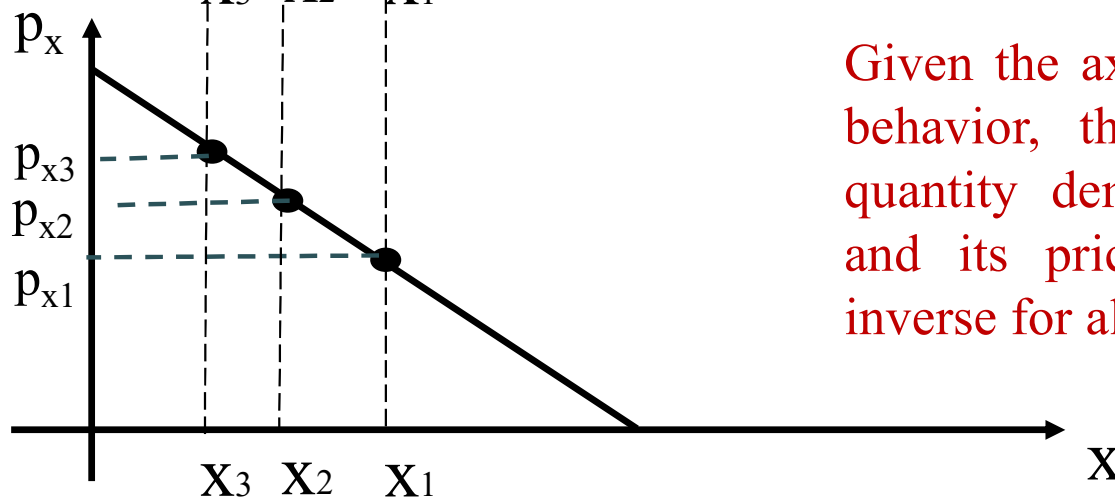
Consumer's equilibrium to consumer's demand



$$\text{Equation } B_1: x \cdot p_{x1} + y \cdot p_y = M$$

$$\text{Equation } B_2: x \cdot p_{x2} + y \cdot p_y = M \quad p_{x2} > p_{x1}$$

$$\text{Equation } B_3: x \cdot p_{x3} + y \cdot p_y = M \quad p_{x3} > p_{x2}$$



Given the axioms of consumers' behavior, the relation between quantity demanded for a good and its price is bound to be inverse for all non-Giffen goods.

Problem 7 – Set 1

A consumer's utility function is given by $U = x + 2y$. Draw the consumer's indifference map. What is her marginal rate of substitution? Suppose her income is 30. Draw her demand curve for x when the price of y is 6. Also draw her demand curve for y when the price of x is 3.

Solution to Problem 7 – Set 1

$$U = x + 2y \quad \text{MRS} = -1/2$$

$$\text{Budget } 30 = p_x x + 6y, \quad \text{Slope} = -p_x / 6$$

BL is steeper than IC if $p_x > 3$. $x = 0$ in eq. DD of $x = 0$

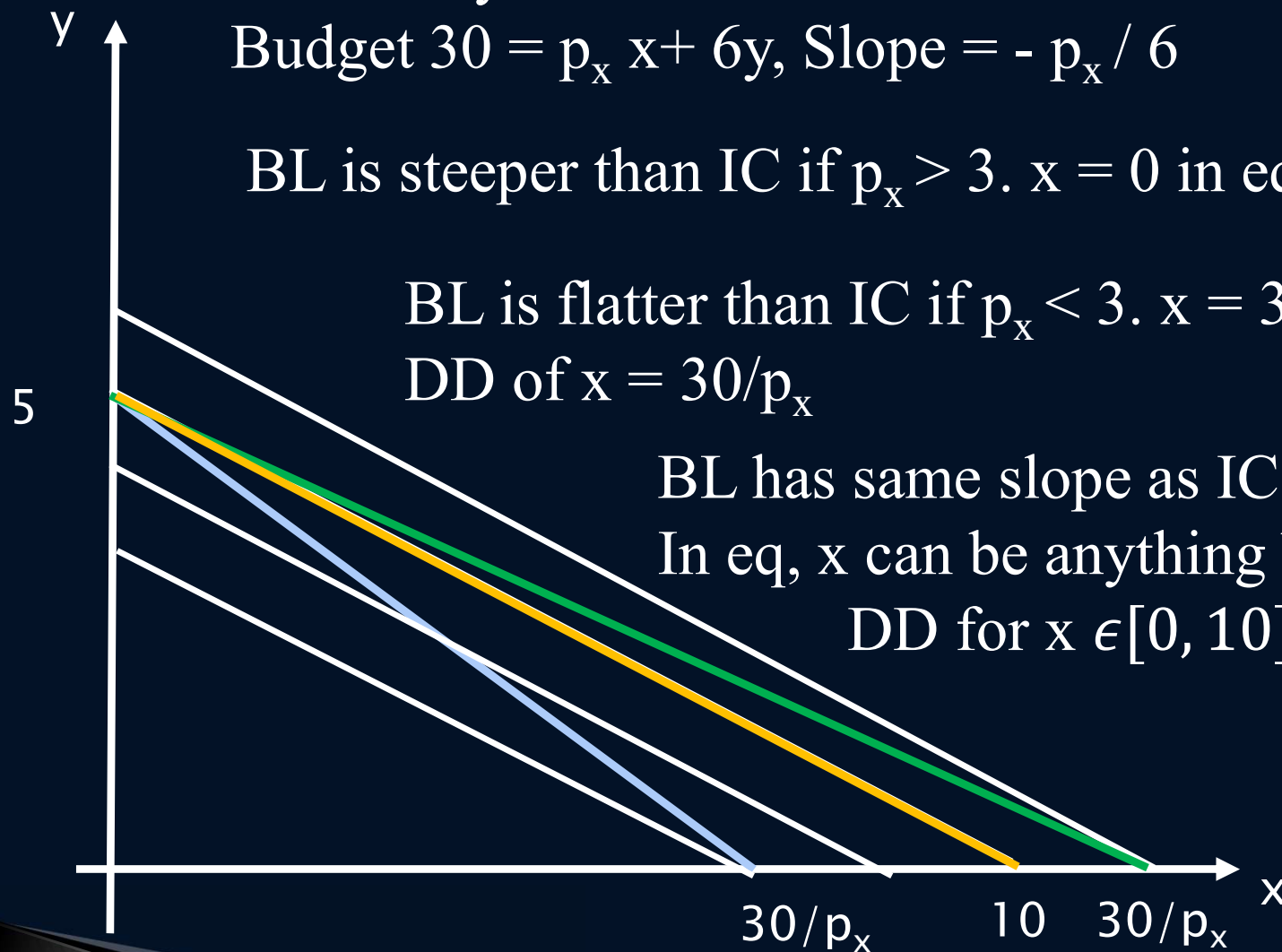
BL is flatter than IC if $p_x < 3$. $x = 30/p_x$ in eq.

DD of $x = 30/p_x$

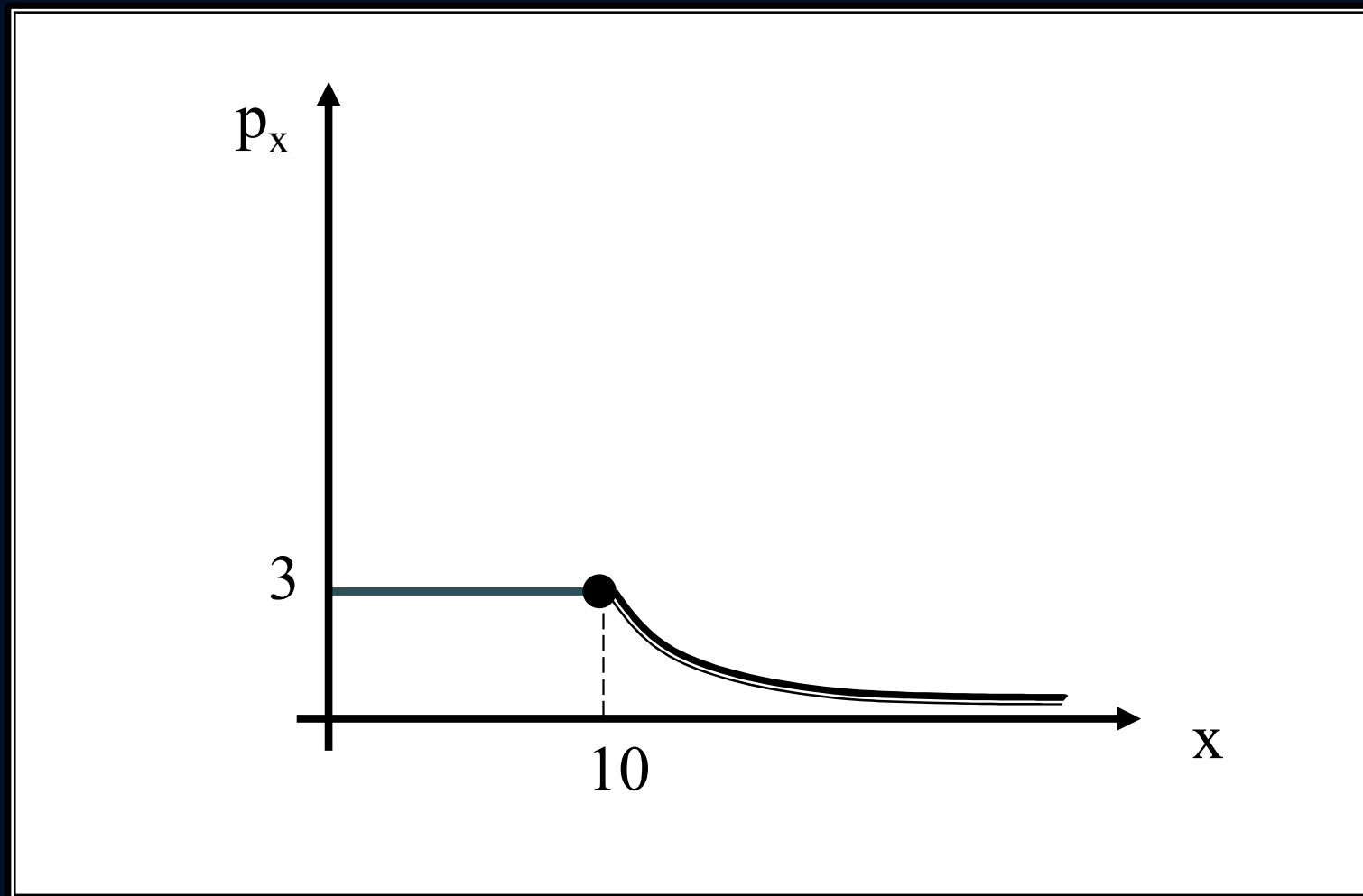
BL has same slope as IC if $p_x = 3$.

In eq, x can be anything between 0 and 10.

DD for $x \in [0, 10]$



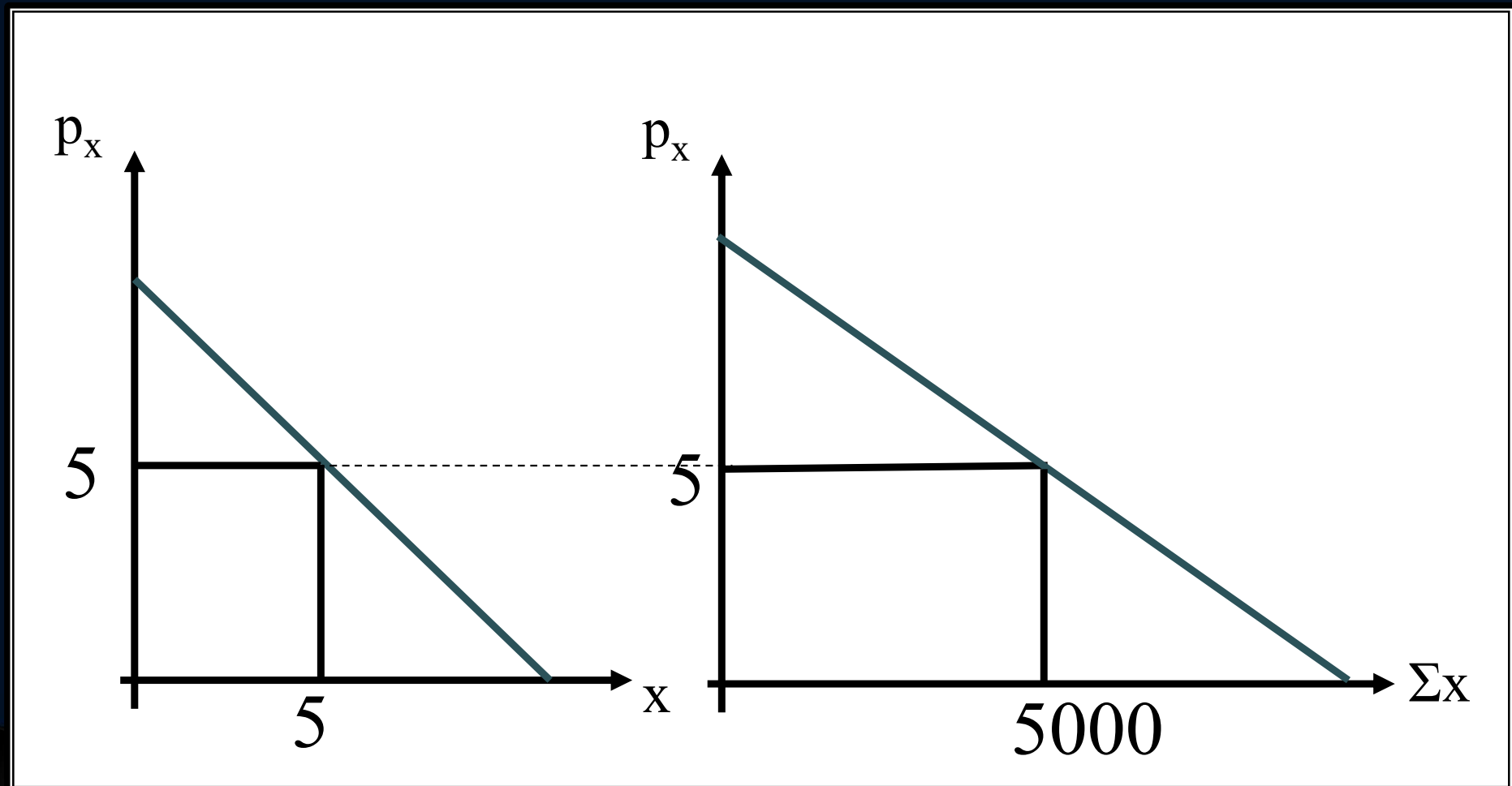
Demand function for x when $p_y = 6$



Derive the demand function for y when $p_x = 3$

Market Demand

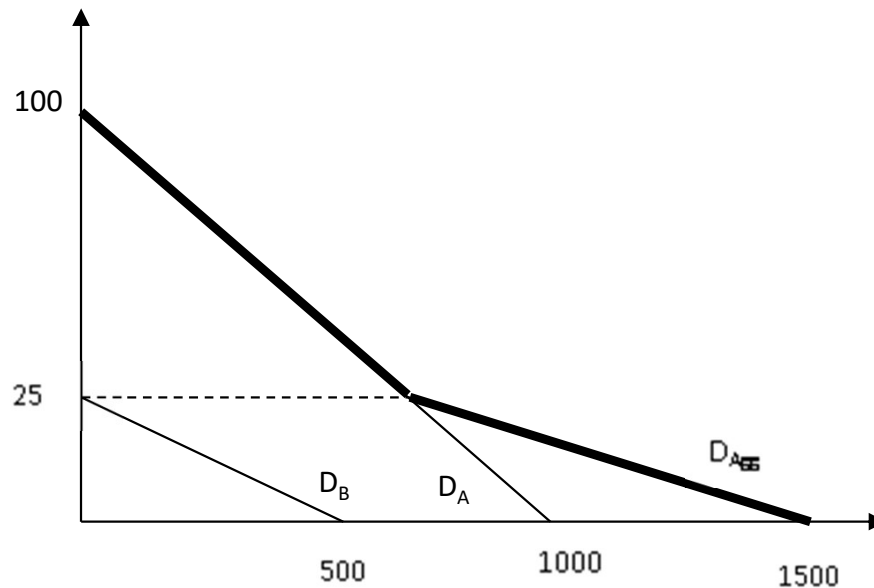
- ▶ Market demand curve is derived by horizontally summing individual demand curves



Aggregating non-homogenous demand fns

- ▶
- ▶ Two consumers (or two market segments) – A and B.
- ▶ $x_A = 1000 - 10P$
- ▶ $x_B = 500 - 20P$
- ▶ What is the aggregate demand function?

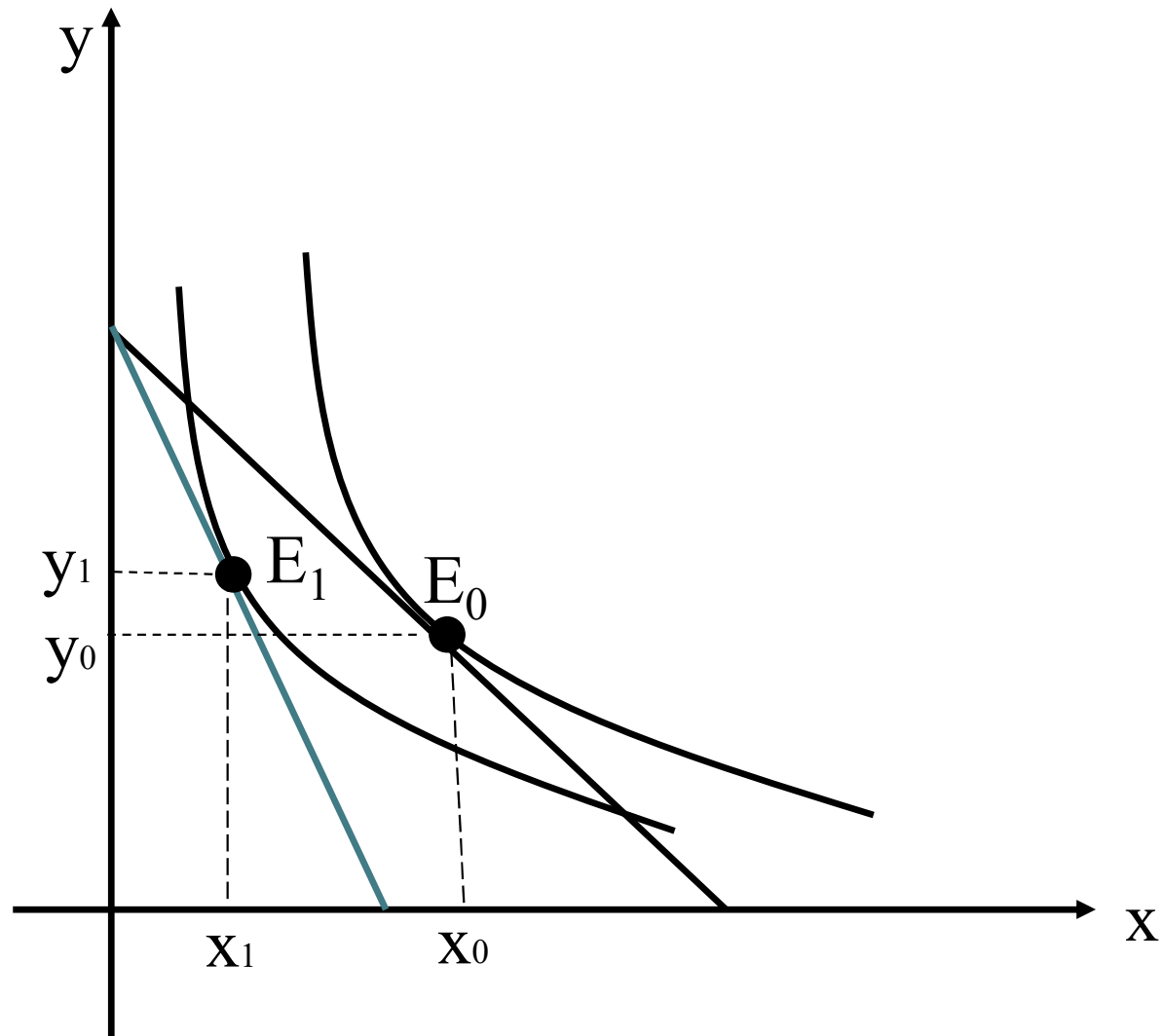
- ▶ The aggregate demand function is:
- ▶ $x = 1500 - 30P$ if, $0 \leq P < 25$ (When $P < 25$ there is demand from both A and B)
- ▶ $= 1000 - 10P$ if, $25 \leq P < 100$ (When $P \geq 25$ demand from B becomes zero.)



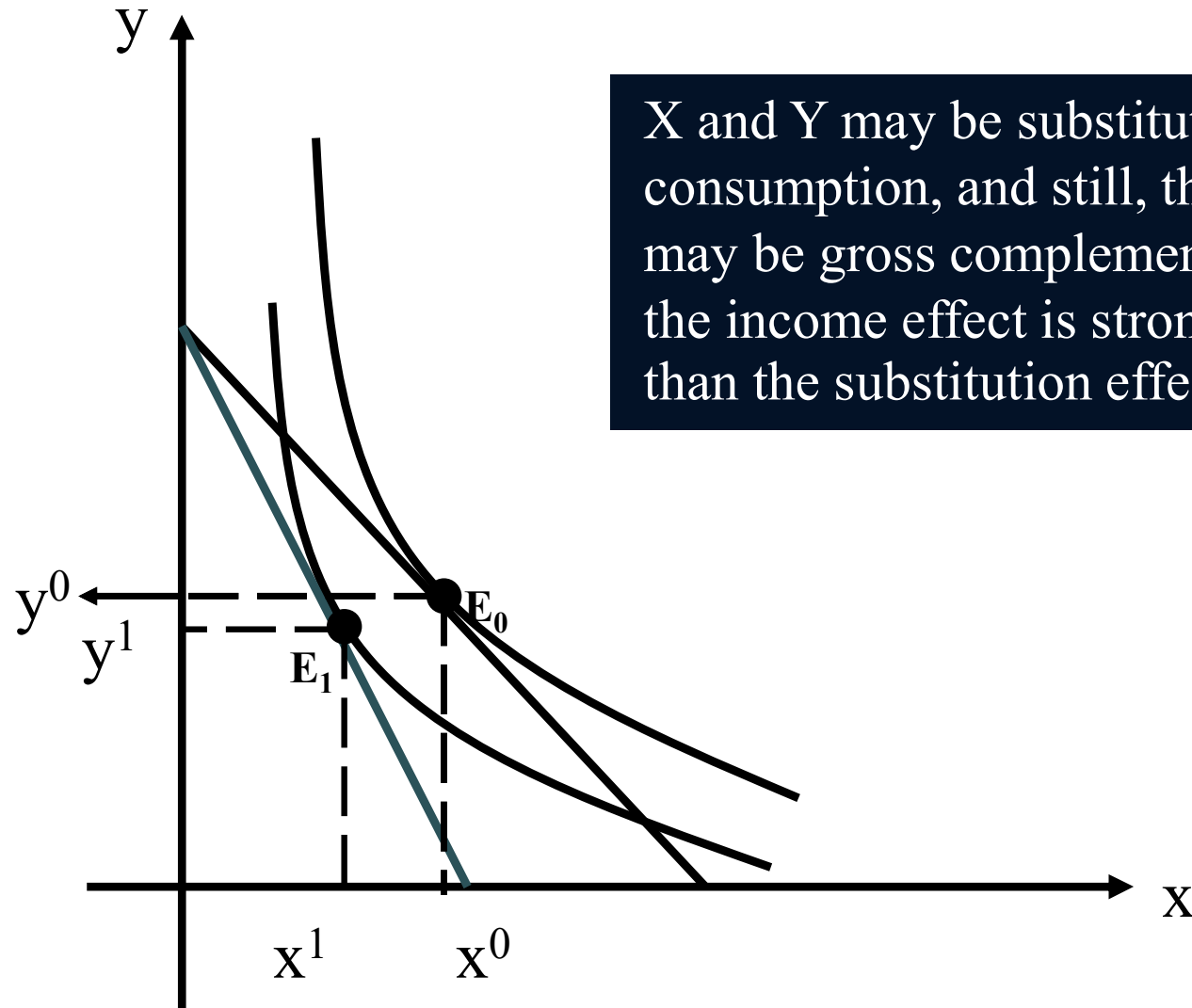
Definitions - Substitutes and complements

- X and Y are gross substitutes if the demand of Y increases with a rise in the price of X.
 - X and Y must be substitutes in consumption.
- X and Y are gross complements if the demand of Y decreases with a rise in the price of X.

X and Y are gross substitutes



X and Y are gross complements

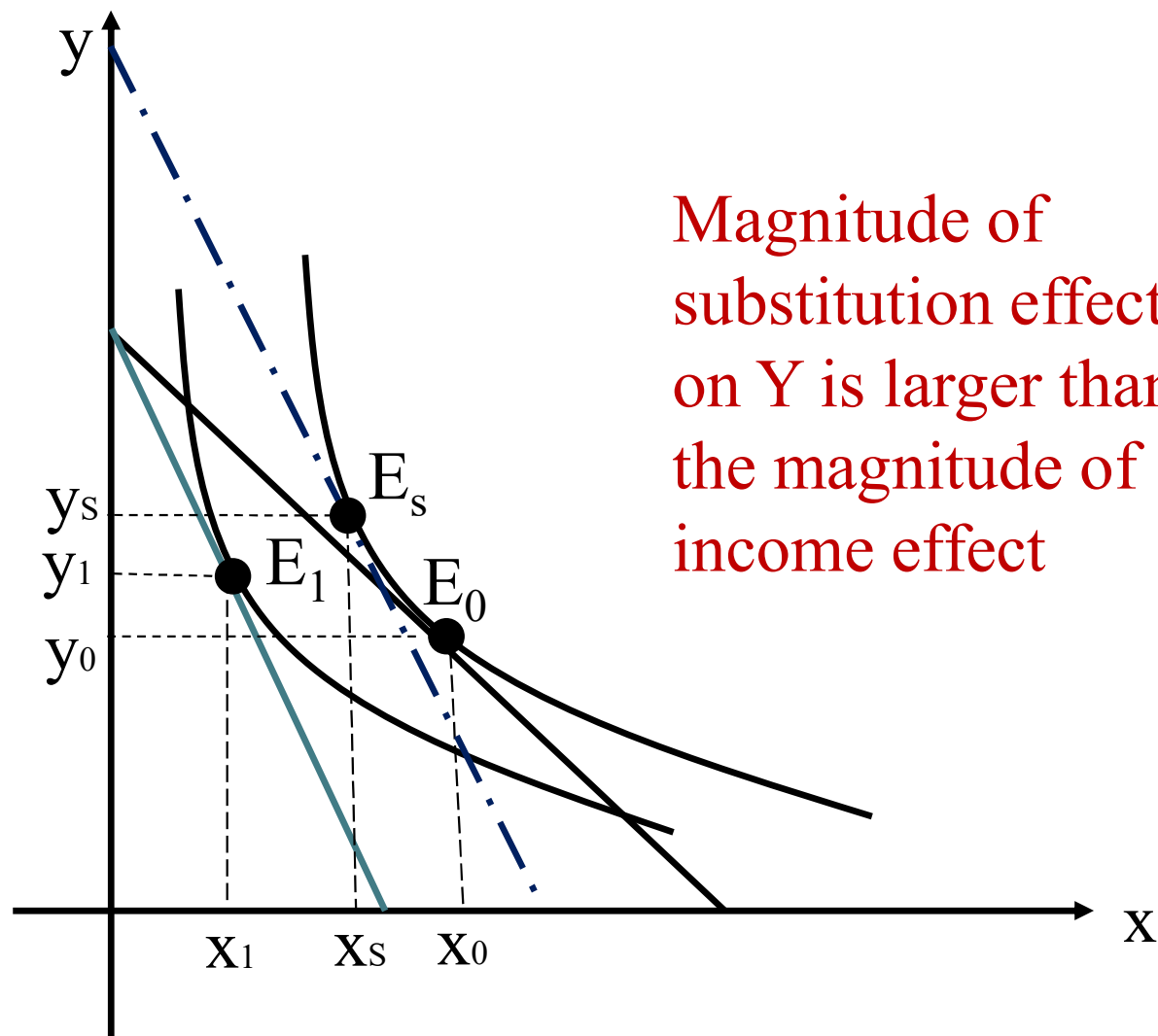


X and Y may be substitutes in consumption, and still, they may be gross complements if the income effect is stronger than the substitution effect.

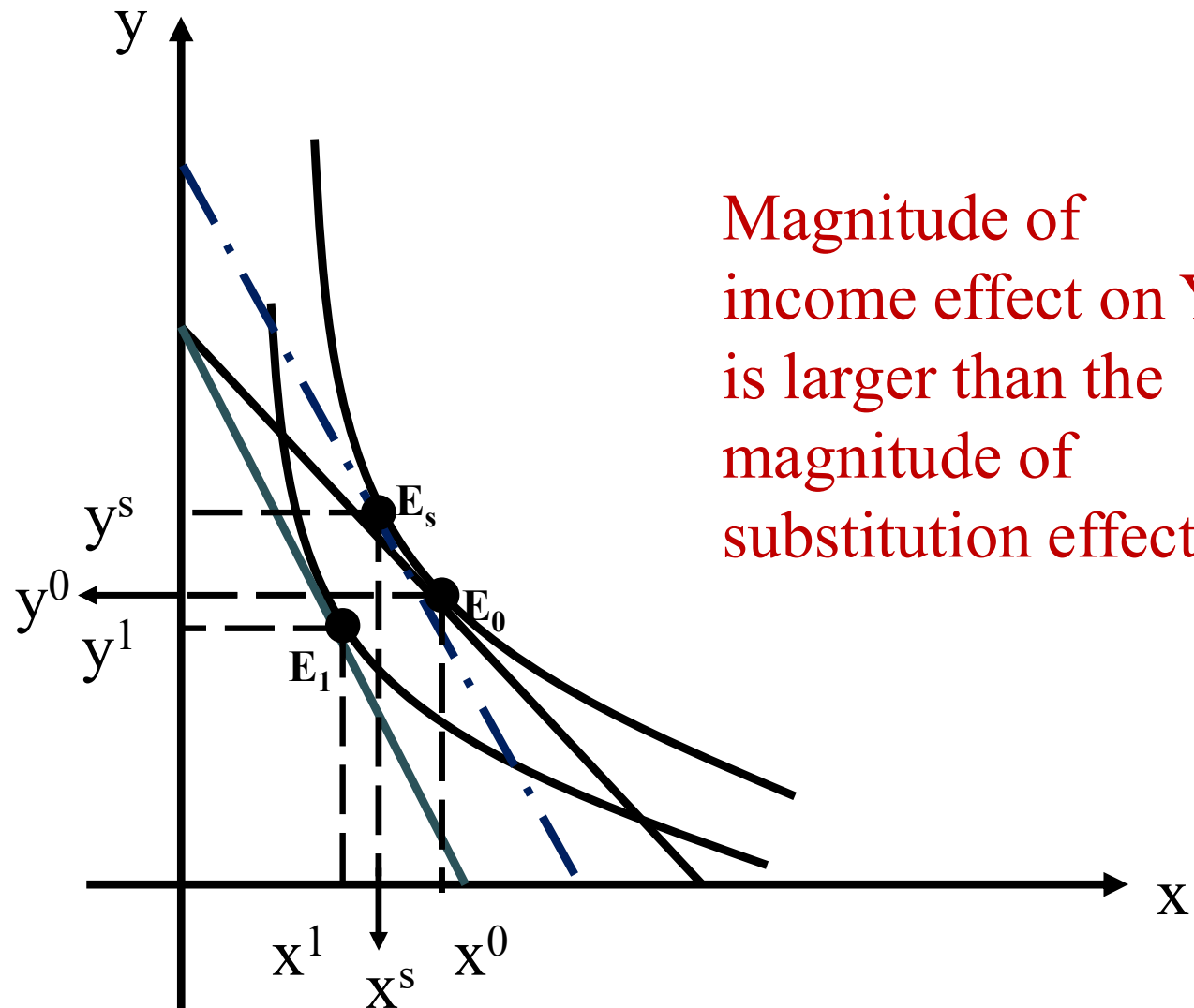
Substitutes and complements – Substitution and income effects

- X and Y are gross substitutes if
 - The magnitude of substitution effect is more than that of income effect.
- X and Y are gross complements if
 - The magnitude of income effect is more than that of substitution effect.
 - If X and Y are unrelated in consumption or complements in consumption, the entire price effect is due to income effect.

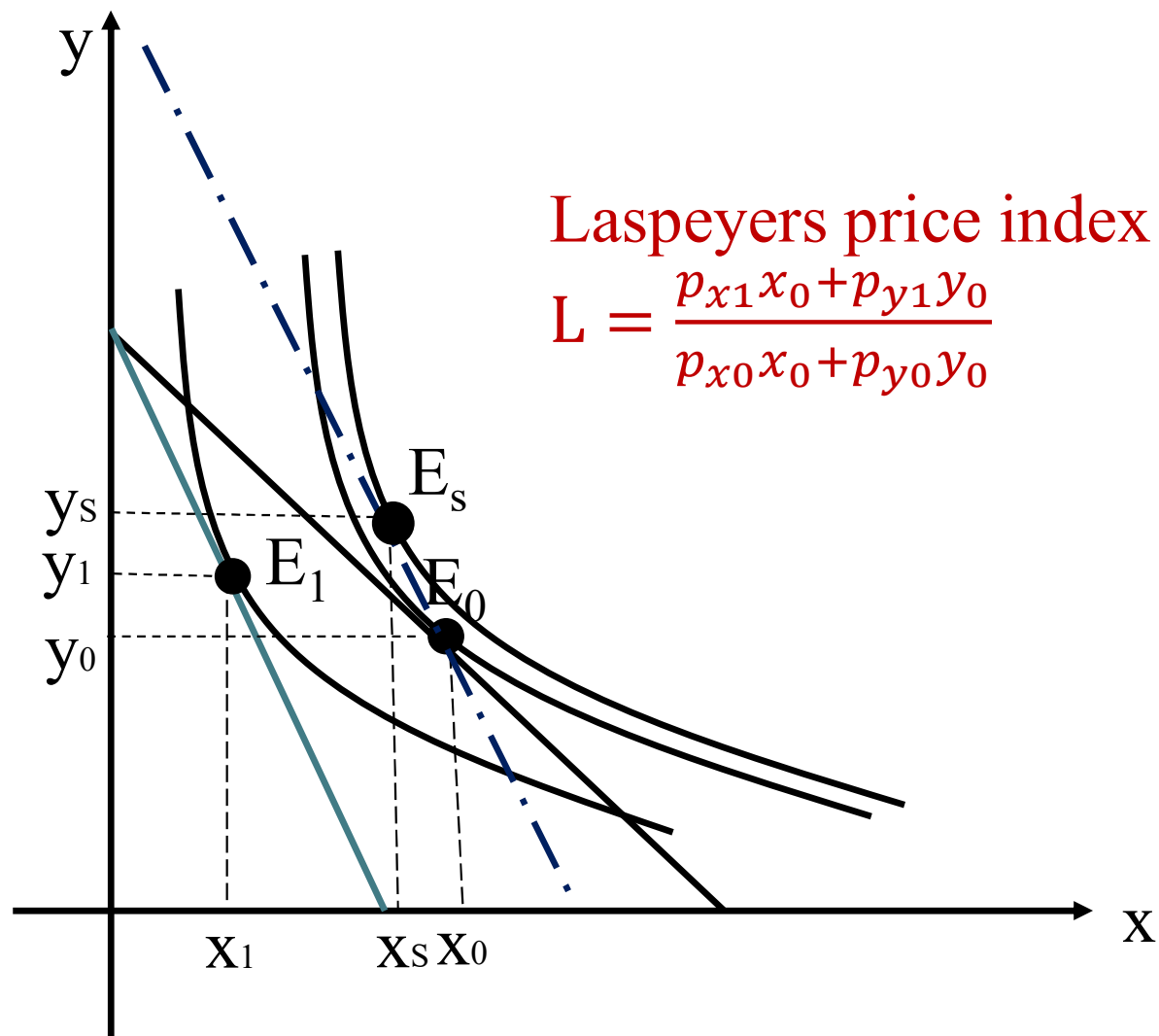
X and Y are gross substitutes



X and Y are gross complements



Compensation by price index



Cross elasticity of demand – a measure of consumer's sensitivity to changes in prices of substitute / complement

- ▶ Cross elasticity of demand captures the responsiveness of demand to changes in price of a related good

$\gamma = (\text{percentage change in demand}) / (\text{percentage change in price of a related good})$

$$= \frac{p_y dx}{x dp_y}$$

- ▶ If Y is a gross substitute of X then $\gamma > 0$
- ▶ If Y is a gross complement of X then $\gamma < 0$

Income elasticity of demand

- ▶ Income elasticity of demand captures the responsiveness of demand to changes in consumer's disposable income
- ▶ $\eta = \frac{Mdx}{xdM}$
- ▶ $\eta \geq 0$ for Normal goods
- ▶ $\eta < 0$ for Inferior goods

Managerial Economics

Production and cost

Sumit Sarkar, XLRI Jamshedpur

Term I

Production Function

- Production Technology is given as a relation between inputs and outputs
- $(x_1, x_2, \dots, x_n) = f(y_1, y_2, \dots, y_m)$
- This function is called Production Function!
- It captures the technology embedded in the production process.

Simple Short-run Production Function

- $x = f(L, K_0)$
 K_0 – fixed capital used
 L – labor units employed
- L is ‘hours or time units of labor’, which is a unit in time and not ‘number of workers’. L can be taken as ‘number of workers’ if and only if all workers are employed for the equal number of hours.
- By definition, the production process is in the **short run** long as the capital deployment of the firm is **fixed**.

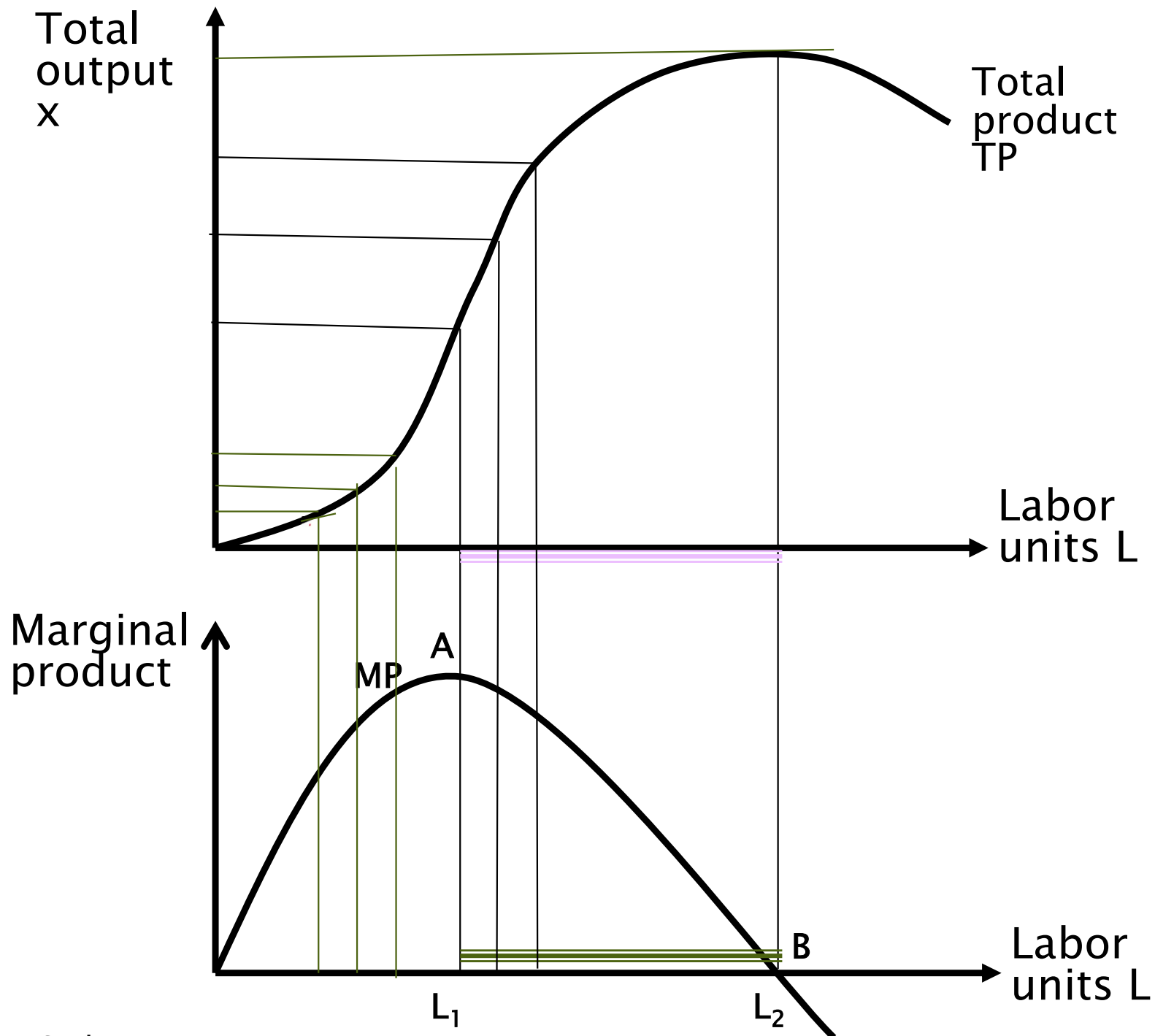
Some definitions

- **Marginal Product (Return to factor):**
Change in output for a very small change in the variable input (labor)

$$MP_L = \delta x / \delta L$$

- **Average Product:**
Output per unit of the variable input (labor)

$$AP_L = x / L$$



Typical Manufacturing Production Fn.

- In zone I, it is increasing and convex i.e., it exhibits increasing returns to factors

$$\delta x / \delta L = MP_L > 0$$

$$\delta^2 x / \delta L^2 = (\delta MP_L / \delta L) > 0$$

- In zone II, it is increasing and concave, i.e., it exhibits diminishing returns to factors

$$\delta x / \delta L = MP_L > 0$$

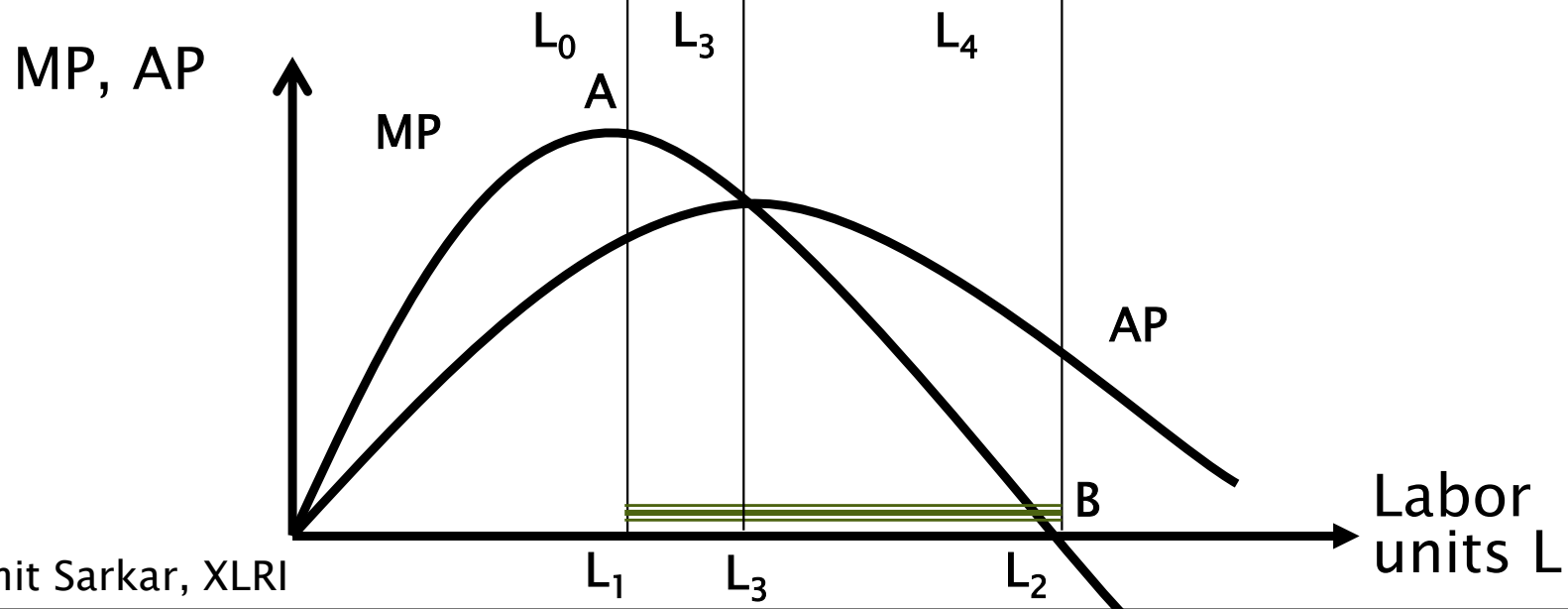
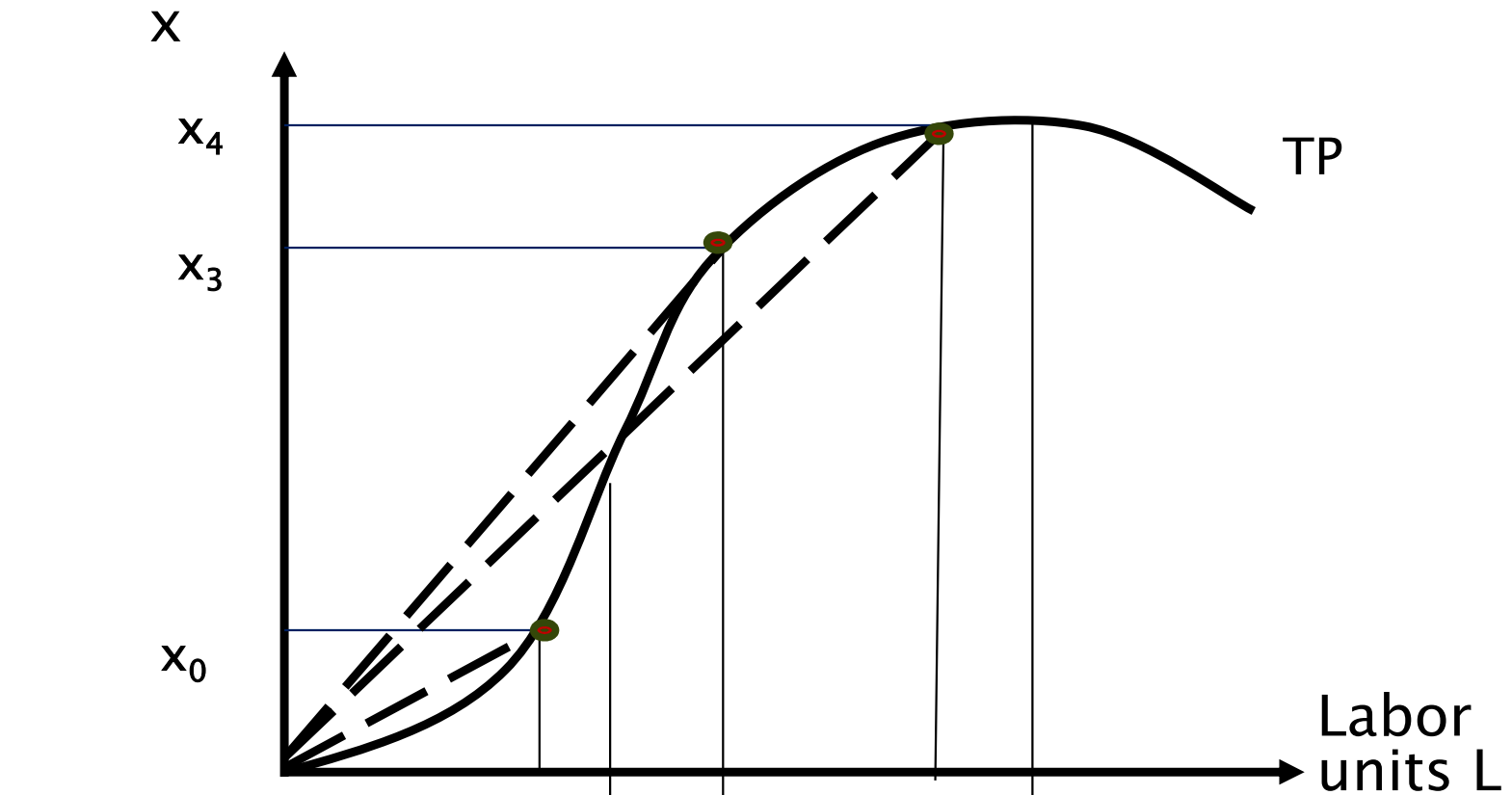
$$\delta^2 x / \delta L^2 = (\delta MP_L / \delta L) < 0$$

- In zone III, it is decreasing i.e., it exhibits negative returns to factors

$$\delta x / \delta L = MP_L < 0$$

Where should the firm produce?

- In zone I?
 - MP is increasing at an increasing rate. If you increase labor employment, your output will increase at a faster rate.
- Certainly not in zone III
 - Output is decreasing
- In zone II
 - Where in zone II?



MP cuts AP at its maxima

$$AP_L = x / L$$

- First order condition for maxima:

$$\delta(x/L) / \delta L = 0$$

$$\text{or, } [(\delta x / \delta L) \cdot L - x] / L^2 = 0$$

$$\text{or, } (\delta x / \delta L) = x / L$$

$$\text{or, } MP_L = AP_L$$

- Second order condition $(\delta MP_L / \delta L) < 0$ i.e., the slope of the MP curve is negative at the maximum point of the AP, i.e., *MP cuts AP from above*.

Suppose L and K are to be used in fixed proportion

Inputs are perfect complements

Example: One worker can produce 10 units working on a machine for 1 hour.

If the plant runs 24x7 and the workers are employed on hourly basis (measure labor in person-hours) how does the short-run production function look like?

A Problem

Sub-zero produces liquid nitrogen. Its production function is given as:

$$Q = 4K^{1/2}L^{1/2}.$$

Sub-zero does not own a plant (capital), but rented in one plant ($K = 1$) and pays Rs. 2000 / day as rent. The raw material required is atmospheric air and is available free. Only other factor of production is labor. If the wage rate is Rs. 160 / day (8 hours make a work day), write down an equation that represents Sub-zero's daily cost as a function of quantity produced.

Production function to cost function

In SR, $K = 1$.

Hence, $Q = 4L^{1/2}$

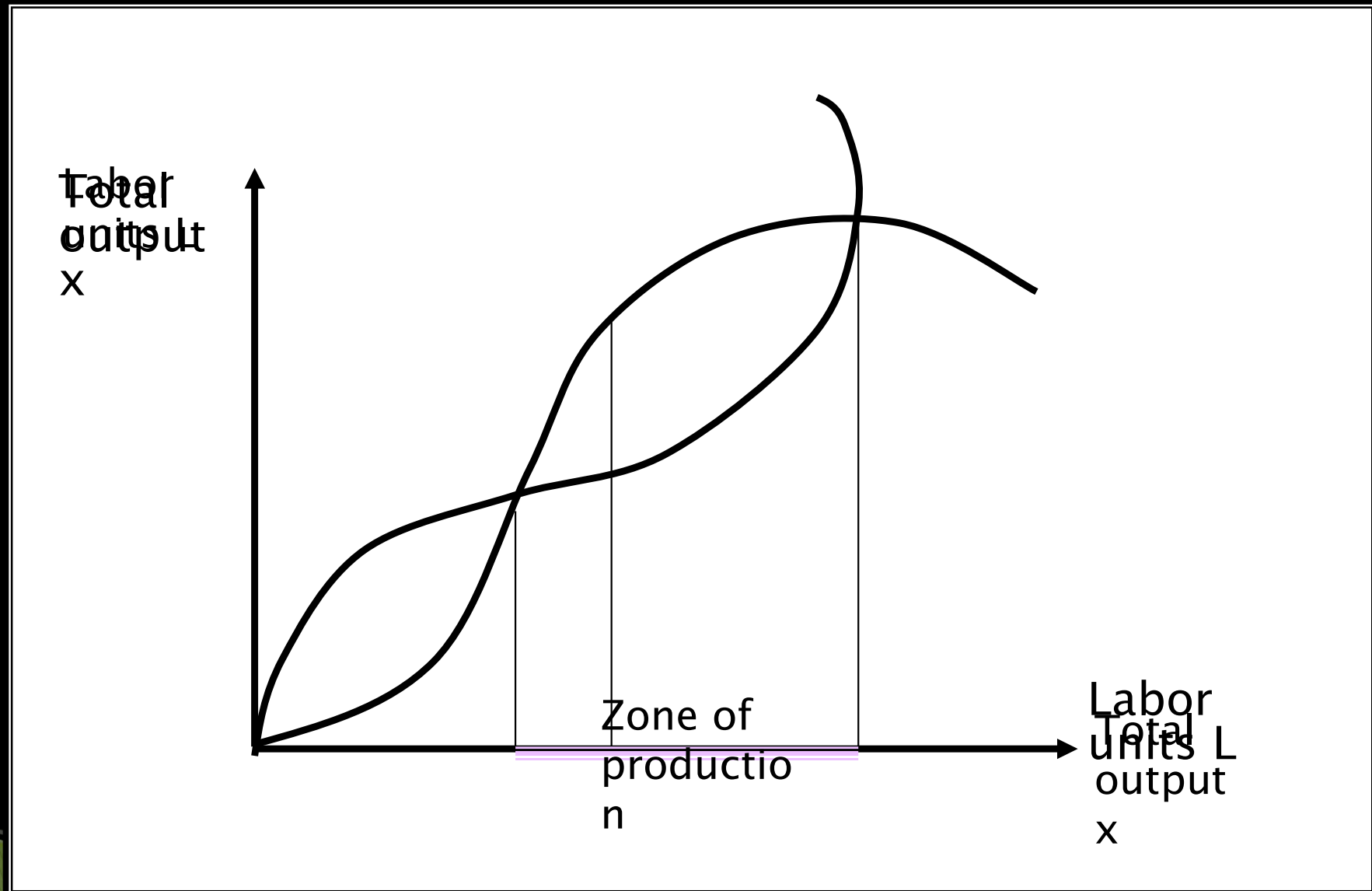
or $L = Q^2/16$

Therefore, $C = r.K + w.L$
 $= 2000 + 160 (Q^2/16)$

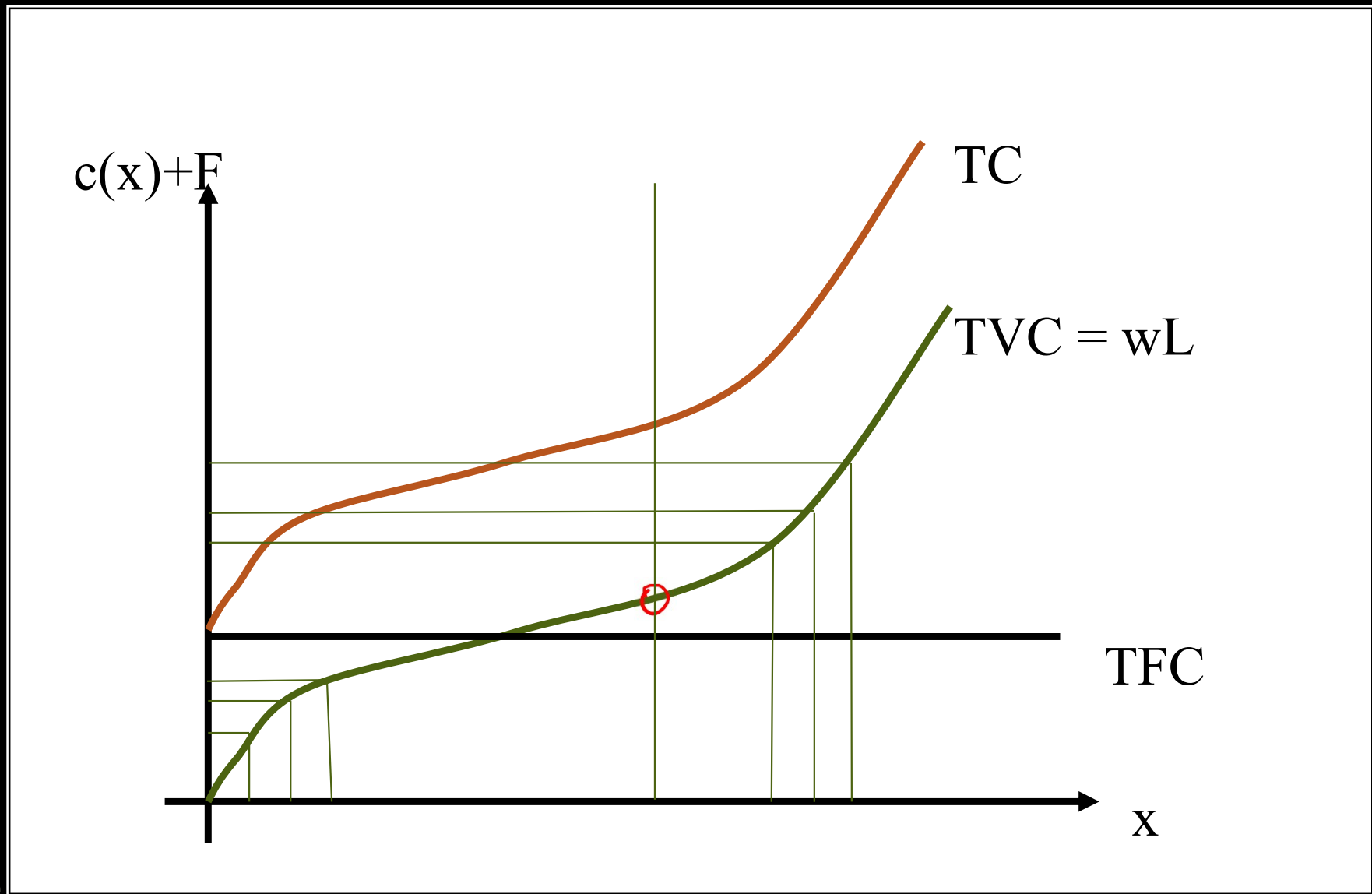
Cost function (daily)

$$C = 2000 + 10Q^2$$

We get variable cost by inverting the production function



Cost Function



Marginal and average costs - definitions

- Marginal Cost:

Change in cost (variable) for a very small change in the output (quantity)

$$MC = dc(x) / dx$$

- Average variable Cost:

Variable cost per unit of output

$$AVC = VC / x$$

- Average fixed cost:

Fixed cost per unit of output

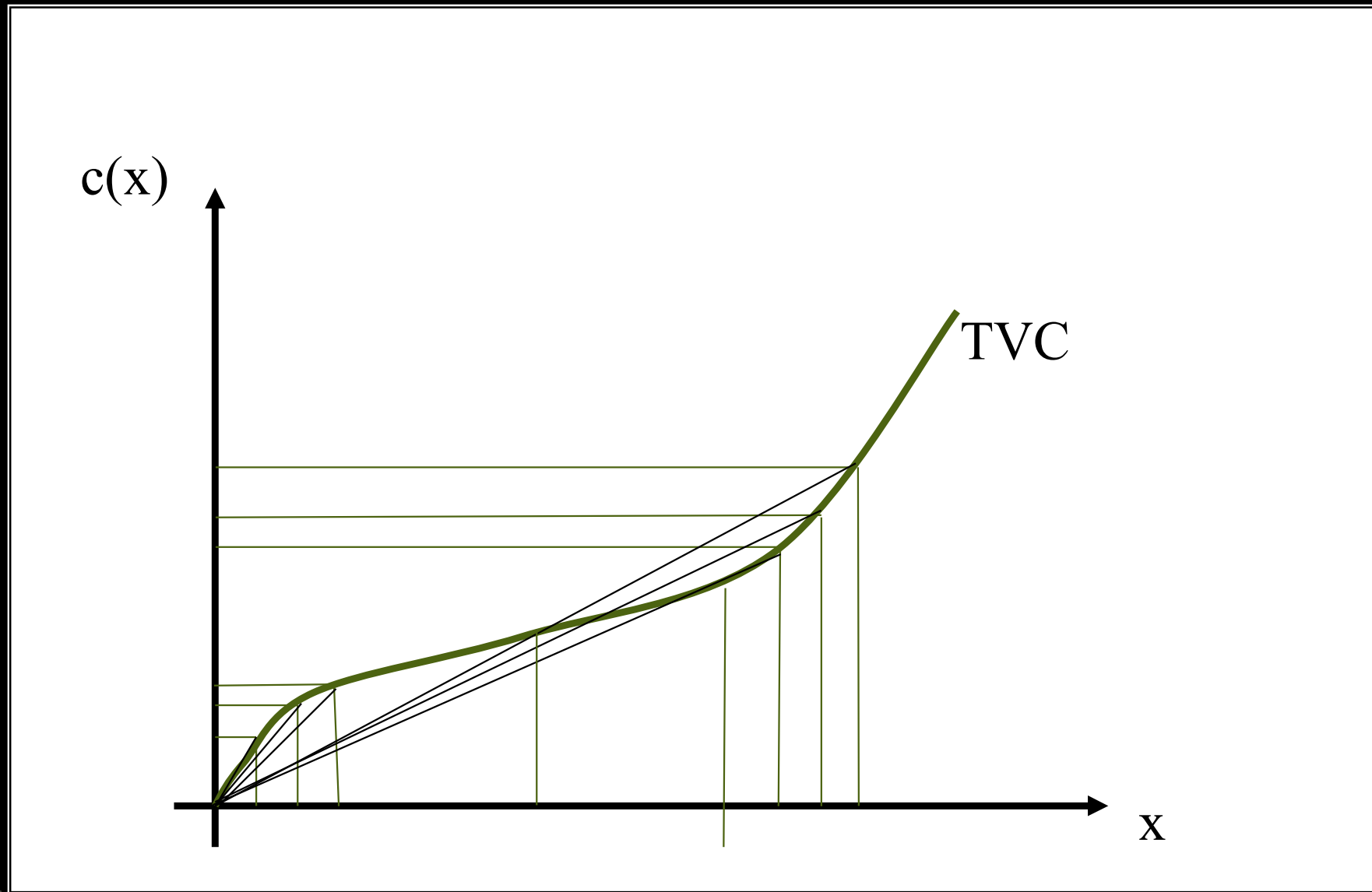
$$AFC = FC / x$$

- Average Cost:

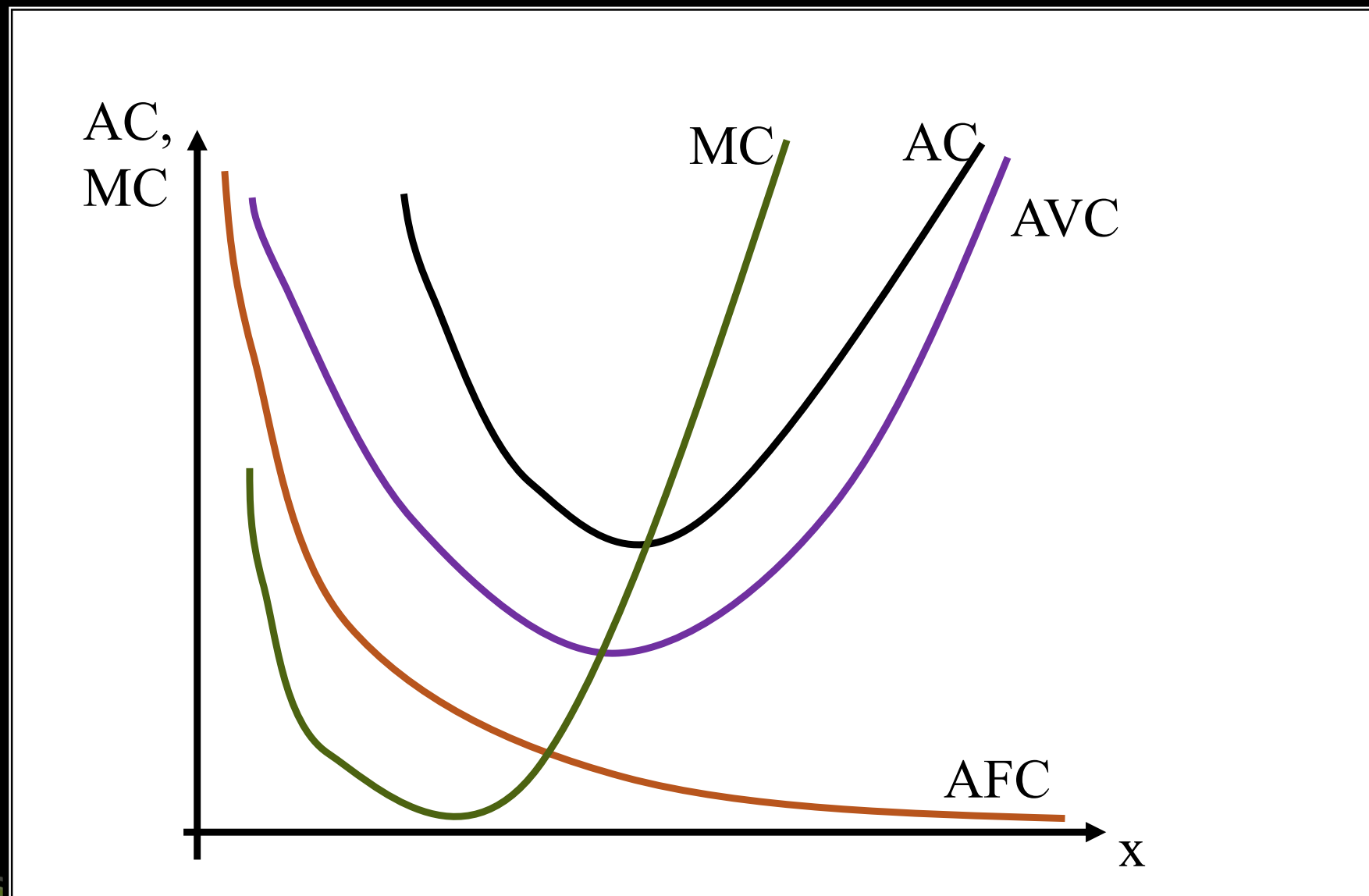
Cost per unit of output

$$AC = TC / x = (FC + VC) / x = AFC + AVC$$

AVC and MC on Variable Cost Function



Short run AC and MC



MC cuts AVC at the minima

$$TC = FC + VC = F + c(x)$$

$$AC = TC / x = [F + c(x)] / x = [F / x] + [c(x) / x]$$

$$AVC = c(x) / x$$

- First order condition for minima:

$$d(AVC) / dx = 0$$

$$\text{or, } [(dc(x) / dx) \cdot x - c(x)] / x^2 = 0$$

$$\text{or, } (dc(x) / dx) = c(x) / x$$

$$\text{or, } MC = AVC$$

- Second order condition

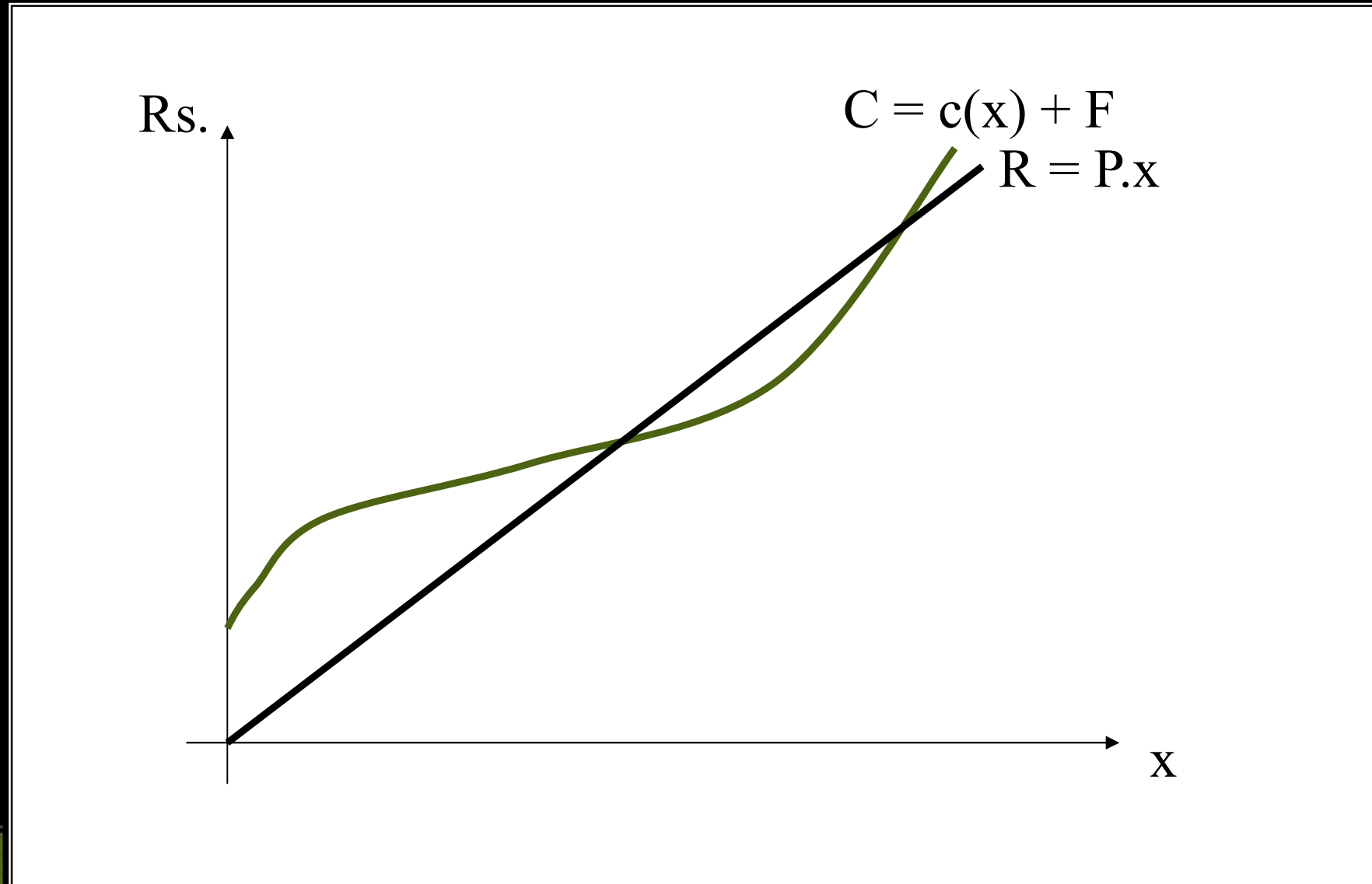
$$d^2(AVC) / dx^2 > 0$$

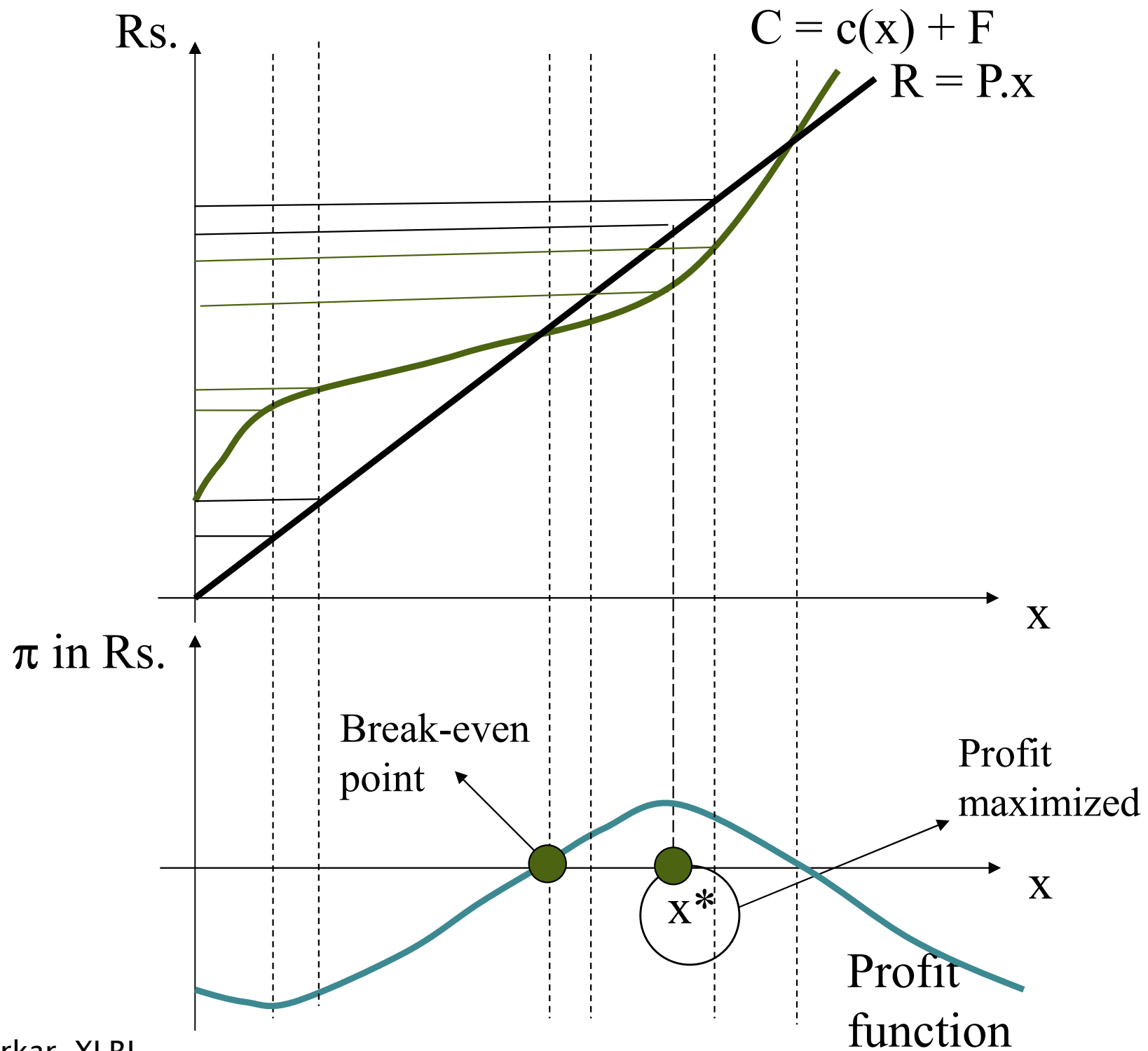
$$dMC / dx > 0$$

- 2nd order condition implies

- The slope of the MC curve is more than the slope of the AVC curve (which is 0 at the minimum point) at the point of intersection, i.e., MC is upward sloping at $AVC_{\min}$

Revenue and cost function – Price-taking firm





Profit maximizing condition for a price-taking firm

Profit function:

$$\pi = P.x - c(x) - F$$

First order condition:

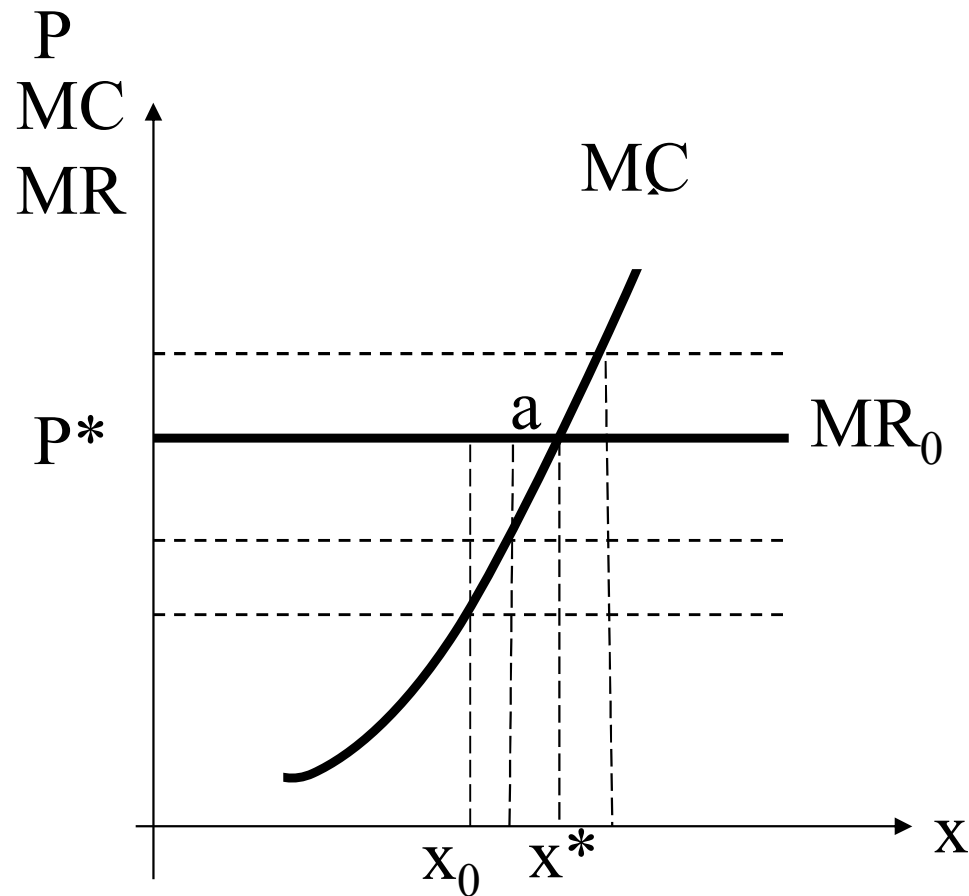
$$P = (MR) = dc(x)/dx = MC$$

Second order condition:

$$d^2c(x)/dx^2 > 0$$

Profit is maximized when the MC is equal to the market price and the MC is increasing.

Why price-taking firm's profit is max at $P=MC$



Managerial Economics (MGEXOB25-1)

Problem Set 1 - Answers

1. Suppose that the market supply and demand curves for wheat are, respectively,

$$P = 0.002 X_s$$

$$P = 30 - 0.001 X_d$$

Prices are in ₹ per KG, and quantities in KG.

Government decided to buy enough wheat retain the price at ₹2 higher than the equilibrium price. The wheat bought by the government will be used in the National Rural Employment Program, which is outside the market system.

How much wheat (in KG) will the government buy?

Done in class

2. Suppose that the government decided NOT to buy any wheat. Instead, they offer a subsidy per KG of wheat sold. How much subsidy (per KG) is required to retain the price at ₹2 higher than the equilibrium price derived in Problem 1?

Which policy is more expensive for the government, the one given in Problem 1 or this one?

Done in class

3. A consumer's utility function is given by $U = x^{1/3}y^{2/3}$.

- Is the utility increasing in x? Explain
- Is the utility increasing in y? Explain
- Find the marginal utility of X as a function of x and y.
- Find the marginal utility of Y as a function of x and y.
- Is the MU of X decreasing? Explain
- Is the MU of Y decreasing? Explain
- Find the MRS when the consumer consumes 100 units of X and 200 units of Y

Done in class

4. A consumer's utility function is given by $U = x^{0.5}y^{0.5}$. What is her $MRS_{X \text{ for } Y}$ when she consumes 5 units of Y and 1 unit of X? Suppose, her income is 30 and the prices of X and Y are 2 and 1 respectively. Is she on her equilibrium? How much X and Y should she consume at her equilibrium?

Done in class

5. Dr. Wooster consumes two goods – coffee and sugar. He consumes exactly two teaspoons of sugar per teaspoon of coffee. Graph Wooster's indifference curves in the coffee – sugar space. If Wooster has \$100 to spend on coffee and sugar, and sugar costs \$0.20 per teaspoon and coffee costs \$0.60 per teaspoon, how much coffee and sugar does he buy?

Done in class

6. A consumer's utility function is given by $U = x + 2y$. Draw the consumer's indifference map. What is her marginal rate of substitution? Suppose her budget is 30, price of x is 4, and the price of y is 6. How much x and y does she consume in the equilibrium?

Done in class

7. A consumer's utility function is given by $U = x + 2y$. Draw the consumer's indifference map. What is her marginal rate of substitution? Suppose her income is 30. Draw her demand curve for x when the price of y is 6. Also draw her demand curve for y when the price of x is 3.

Done in class

Practice questions:

(Type of questions you may expect in Quiz 1 – Answers will be provided on 13th Sept, after you try out on your own during the week following the classes on 7th Sept)

1. A consumer's utility function is $U = 2x + 3y$, for $x \geq 0, y \geq 0$. Her budget for x and y is 100, and prices of x and y are 3 and 4, respectively. Find her equilibrium consumption bundle.

Ans. (0, 25)

2. A consumer consumes x and y in the ratio 9:11. Her budget for x and y is 198, and prices of x and y are 1 and (9/11), respectively. Find her equilibrium consumption bundle.

Ans. (99, 121)

3. A consumer's utility function is

$$U = 4\sqrt{xy}$$

for $x \geq 0, y \geq 0$. Her budget for x and y is 60, and the prices of x and y are 3 and 5, respectively. Find her equilibrium consumption bundle.

Ans. (10, 6)

4. A consumer's utility function is

$$U = x^{1/3}y^{1/3}$$

for $x \geq 0, y \geq 0$. If she consumes equal quantities of x and y , then what is her $MRS_{(x \text{ for } y)}$?

Ans. -1

5. A consumer consumes 3 goods, x , y and z , and her utility function follows 'more is better' and the 'law of DMU' in all 3 goods. The prices of x , y and z are 10, 15 and 20 respectively. In her equilibrium, MU of x is 0.5 and MU of z is 1. How much is her MU of y in the same equilibrium?

Apply the law of equi-marginal utility.

$$MU / \text{price} = 0.5 / 10 = 1 / 20$$

$$\text{Therefore, } MU_y = 15 / 20 = \frac{3}{4}$$

6. When prices of x and y were 2 and 5 respectively, a consumer who perceives x and y as perfect substitutes consumed only x . The consumer stopped consuming x and switched to y due to a rise in the price of x from 2 to 3. There was no change in the price of y . Find her $MRS_{(x \text{ for } y)}$.

Price ratio changes from $2/5$ to $3/5$ and consumption shifts from x to y . Possible only if the slope of IC is between $2/5$ and $3/5$. Hence MRS is in $(-3/5, -2/5)$.

7. The demand for x is given by the function $p = 10 - 0.5x$, where p is the price and x is the quantity. Find the price elasticity of demand when $p = 1$.

Ans. -1/9

8. A consumer consumes x and y in the ratio 2:3. Her budget for x and y is M , and prices of x and y are p and 4, respectively. Determine the consumer's demand function for x .

Here, x and y are perfect complements and $\frac{x}{y} = \frac{2}{3}$. Therefore, in equilibrium, $y = \frac{3}{2}x$. The budget must be satisfied in the eqm. Therefore, $M = px + 4y = (p + 6)x$. Hence, the equilibrium demand for x is: $x = \frac{M}{p+6}$.

9. A consumer's demand function for x is $p = 10 - x$, where p is the price and x is the quantity. If she buys x at $p = 7$, how much is her consumer surplus?

Ans. 4.5

10. The demand for x of a consumer is given by the function,

$$x = \frac{10}{Mp}$$

where M is her income, and p is the price of x . Find the income elasticity of the consumer's demand when $p = 10$

$$\text{Income elasticity} = \left(\frac{dx}{dM} \right) \cdot \left(\frac{M}{x} \right)$$

$p = 10$. Hence, the demand function is: $x = \frac{1}{M}$

$$\frac{dx}{dM} = -\frac{1}{M^2}$$

$$\text{Therefore, income elasticity} = \left(-\frac{1}{M^2} \right) * \left(\frac{M}{x} \right)$$

$$\text{Substituting } x = 1/M \text{ we get, income elasticity} = \left(-\frac{1}{M^2} \right) * (M^2) = -1.$$

Managerial Economics (MGEXOB25-1)

Problem Set 1

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How much wheat (in KG) will the government buy?

Done in class on 1st Sept

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Which policy is more expensive for the government, the one given in Problem 1 or this one?

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- Find the MRS when the consumer consumes 100 units of X and 200 units of Y

To be done in class on 7th Sept

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To be done in class on 7th Sept

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To be done in class on 7th Sept

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To be done in class on 7th Sept

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10. The demand for x of a consumer is given by the function,

$$x = \frac{10}{Mp}$$

where M is her income, and p is the price of x . Find the income elasticity of the consumer's demand when $p = 10$

Managerial Economics (MGEXOB25-1)

Problem Set 2 – Production and Cost

1. Bat-co produces batteries. Bat-co has a production process $f(K,L) = 2K^{1/2}L^{1/2}$. In the short run, only 4 units of capital are available. Write down an equation that represents Bat-co's total costs as a function of quantity, the rental rate (r) and the wage rate (w). If r is \$400/unit and w is \$4/unit, how much does it cost to produce 100 units?

2. A firm's production function is $x = (K^2)(L^{0.5})$. Determine the marginal product (MP) of L when 4 units of L are employed.

3. The cost function of Firm A is $C = 4 + x^2/4$, where x is its output. Determine the MC, AVC, and AFC functions.

4. A firm's production process is characterized by the short-run production function $x = K_0\sqrt{L}$, where $K_0 = 3$. If $w = 27$ and $r = 100$, determine the firm's marginal cost function.

5. The production function of Firm A is

$$x = 5L^{\frac{1}{3}}K^{\frac{2}{3}}$$

The firm uses 3 units of K . The factor prices of K and L are, respectively,

$$r = 1000 \text{ and } w = 250$$

Determine the MC of the firm.

6. Do the following production processes exhibit increasing, decreasing or constant returns to scale in the long run?

a. $f(K,L) = K^{1/2}L^{3/5}$

b. $f(K,L) = 7K^{1/4}L^{3/4}$

c. $f(K,L) = 7K^{1/4}L^{1/2}$

d. $f(K,L) = 14K^2L$

Long Problems

1. Reverse Swing – A caselet in short-run production

Reverse Swing, produces cricket balls in order to meet the demand of the local children who aspire to become cricketers. The production process (with the existing technology) can be represented as: $Q = 160KL$, where Q is output per month, K is capital (machine sheds) used per month and L is workers employed per month (each worker works for 8 hours per day for 25 days in a month). Reverse Swing does not own a machine shed but rents one. Apart from machine shed and labor, Reverse Swing needs 150 grams of cork, 8 square inches of dyed leather and 8 meters of thread to produce each cricket ball. The firm procures the raw materials from competitive market at given prices.

Before starting the business, Reverse Swing had the following information.

- In the community, there are 2000 groups of children who play cricket. They play with cricket balls as well as with tennis balls. The utility function is same across groups and is given as: $U(x,s) = 6x^{1/3}s^{2/3}$, where s represents cricket balls and x represents tennis balls. Each group has Rs. 1200 per month to spend on the two kinds of balls they play with. The market for tennis balls is perfectly competitive and the market price is Rs. 25.
- There are 9 other manufacturers of cricket balls. All the manufacturers use the same technology, purchases the raw materials from the same markets at the same prices, hires labor from the same market at same wage rate and each of them rents in a machine shed @ Rs. 40,000 / month. All the firms produce identical balls. In other words, the manufacturers are identical in all respects and hence equally share the market. In order to produce cricket balls, the producers must obtain a license by becoming a member of the Cricket Balls Manufacturer's Association at a monthly membership fee of Rs. 8000. The association decides the selling price of cricket balls and the members are supposed to accept that price. At present the association fixed the price of cricket balls at Rs. 100.
 - a) Use the utility function, income and prices to calculate how many balls Reverse Swing needs to produce per month.
 - b) Presently Reverse Swing rents only one machine shed. In order to produce enough to meet its own share of monthly market demand how many workers must Reverse Swing employ per month?
 - c) The market price of cork is Rs. 60 / K.G., of leather is Rs. 216 per square foot, of thread is Rs. 0.5 per meter, and the market wage rate is Rs. 4000 per month. Distribution cost is borne by Cricket Balls Manufacturer's Association. What is Reverse Swing's cost per ball?
 - d) What is Reverse Swing's variable cost per ball?
 - e) Write down Reverse Swing's cost function.
 - f) Draw Reverse Swing's marginal cost function.

- g) A market research shows that the number of groups of children playing cricket increased from 2000 to 2500. How many cricket balls will you have to produce now, if all other things in your information set remain unchanged?
- h) How does your labor employment change in the short run (no change in capital deployment)?

2. Cost functions for Gunny

Gunny manufactures jute bags. To produce bags one only needs to stitch up jute fabric using sewing machines. One worker working on a sewing machine can stitch 100 bags in a shift (8 hours). To make a bag 1 metre of jute fabric and 2 metres of jute cord is required. *Gunny* purchases jute fabric @ Rs. 8 a metre, jute cords @ Re. 1 a metre and pays the workers @ Rs. 340 per shift. *Gunny* owns the premises where the gunny bags are produced and stocked, but hires sewing machines @ Rs. 960 per day. Opportunity cost of the premises is zero and it can house 10 machines and 10 workers (1 worker per machine) at a time. If they hire a machine, they have to hire it at least for a day (24 hours). If they employ a worker they will have to employ the worker for at least a shift.

Draw the average cost function and marginal cost function of *Gunny*.

Managerial Economics (MGEXOB25-1)

Problem Set 2 – Production and Cost - **Answers**

1. Bat-co produces batteries. Bat-co has a production process $f(K,L) = 2K^{1/2}L^{1/2}$. In the short run, only 4 units of capital are available. Write down an equation that represents Bat-co's total costs as a function of quantity, the rental rate (r) and the wage rate (w). If r is \$400/unit and w is \$4/unit, how much does it cost to produce 100 units?

$$C = w \frac{x^2}{16} + 4r = \frac{x^2}{4} + 1600$$

2. A firm's production function is $x = (K^2)(L^{0.5})$. Determine the marginal product (MP) of L when 4 units of L are employed.

$$\frac{K^2}{4}$$

3. The cost function of Firm A is $C = 4 + x^2/4$, where x is its output. Determine the MC, AVC, and AFC functions.

$$MC = \frac{x}{2} \quad AVC = \frac{x}{4} \quad AFC = \frac{4}{x}$$

4. A firm's production process is characterized by the short-run production function $x = K_0\sqrt{L}$, where $K_0 = 3$. If $w = 27$ and $r = 100$, determine the firm's marginal cost function.

$$MC = 6x$$

5. The production function of Firm A is

$$x = 5L^{\frac{1}{3}}K^{\frac{2}{3}}$$

The firm uses 3 units of K . The factor prices of K and L are, respectively,

$$r = 1000 \text{ and } w = 250$$

Determine the MC of the firm.

$$MC = \frac{2x^2}{3}$$

6. Do the following production processes exhibit increasing, decreasing or constant returns to scale in the long run?

a. $f(K,L) = K^{1/2}L^{3/5}$ **IRS**

b. $f(K,L) = 7K^{1/4}L^{3/4}$ **CRS**

c. $f(K,L) = 7K^{1/4}L^{1/2}$ **DRS**

d. $f(K,L) = 14K^2L$ **IRS**